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(54) **MULTI-SHIELD SPINAL ACCESS SYSTEM**

(71) Applicant: **MEDOS INTERNATIONAL SARL**,
Le Locle (CH)

(72) Inventors: **Daniel Thommen**, Liestal (CH); **Joern Richter**, Kandern (DE); **Michael Wang**, North Miami, FL (US); **Richard Assaker**, Kain (BE); **Richard Fessler**, Winnetka, IL (US); **Christoph Mehren**, Munich (DE); **Seang Beng Tan**, Clementi (SG); **William Taylor**, Del Mar, CA (US); **Eric Buehlmann**, Duxbury, MA (US); **Michael White**, Liestal (CH)

(73) Assignee: **Medos International Sàrl**, Le Locle (CH)

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See application file for complete search history.

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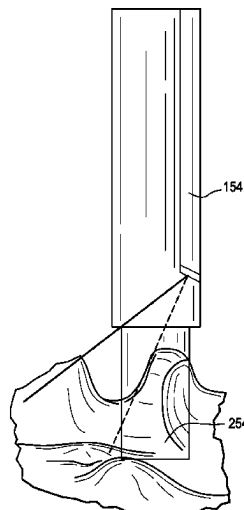
Primary Examiner — Andrew Yang

(74) *Attorney, Agent, or Firm* — Barnes & Thornburg LLP

(57) **ABSTRACT**

An access device for accessing an intervertebral disc having an outer shield comprising an access shield with a larger diameter (~16-30 mm) that reaches from the skin down to the facet line, with an inner shield having a second smaller diameter (~5-12 mm) extending past the access shield and reaches down to the disc level. This combines the benefits of the direct visual microsurgical/mini open approaches and the percutaneous, “ultra-MIS” techniques.

19 Claims, 36 Drawing Sheets



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Chinese Second Office Action for Application No. CN 202080035605.8 dated May 23, 2024 (9 pages).

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Japanese Notice of Reasons for Refusal for Application No. JP 2024-001761 dated Feb. 10, 2025 (8 pages).

FIG. 1A

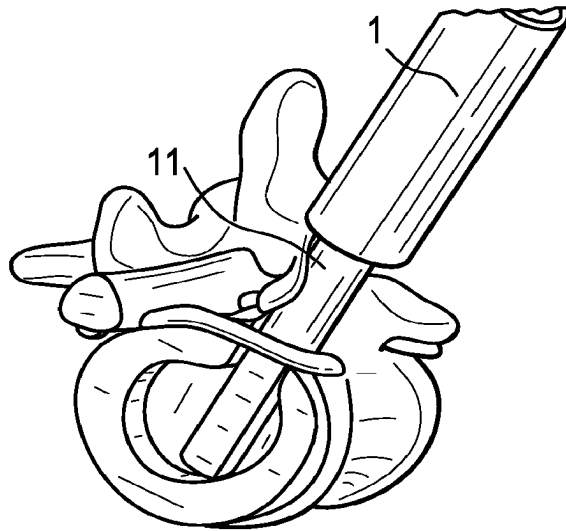


FIG. 1B

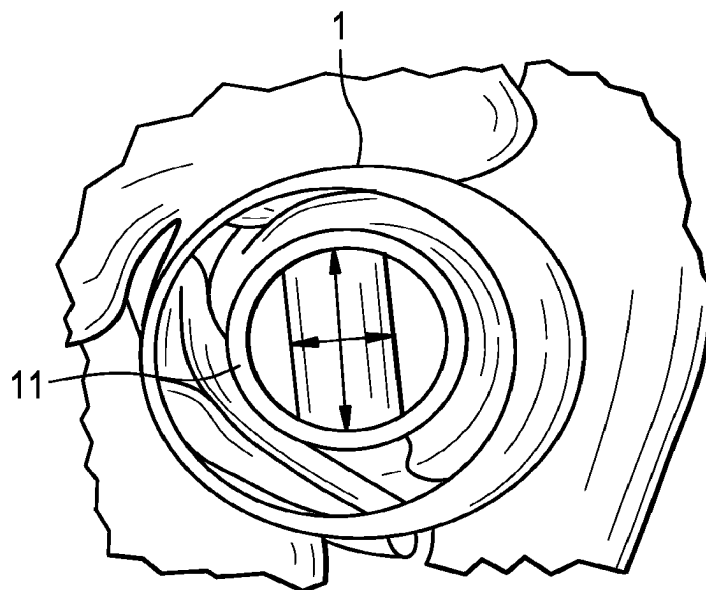


FIG. 2A

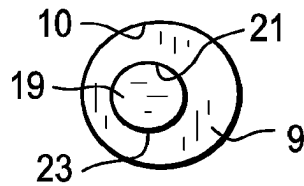


FIG. 2B

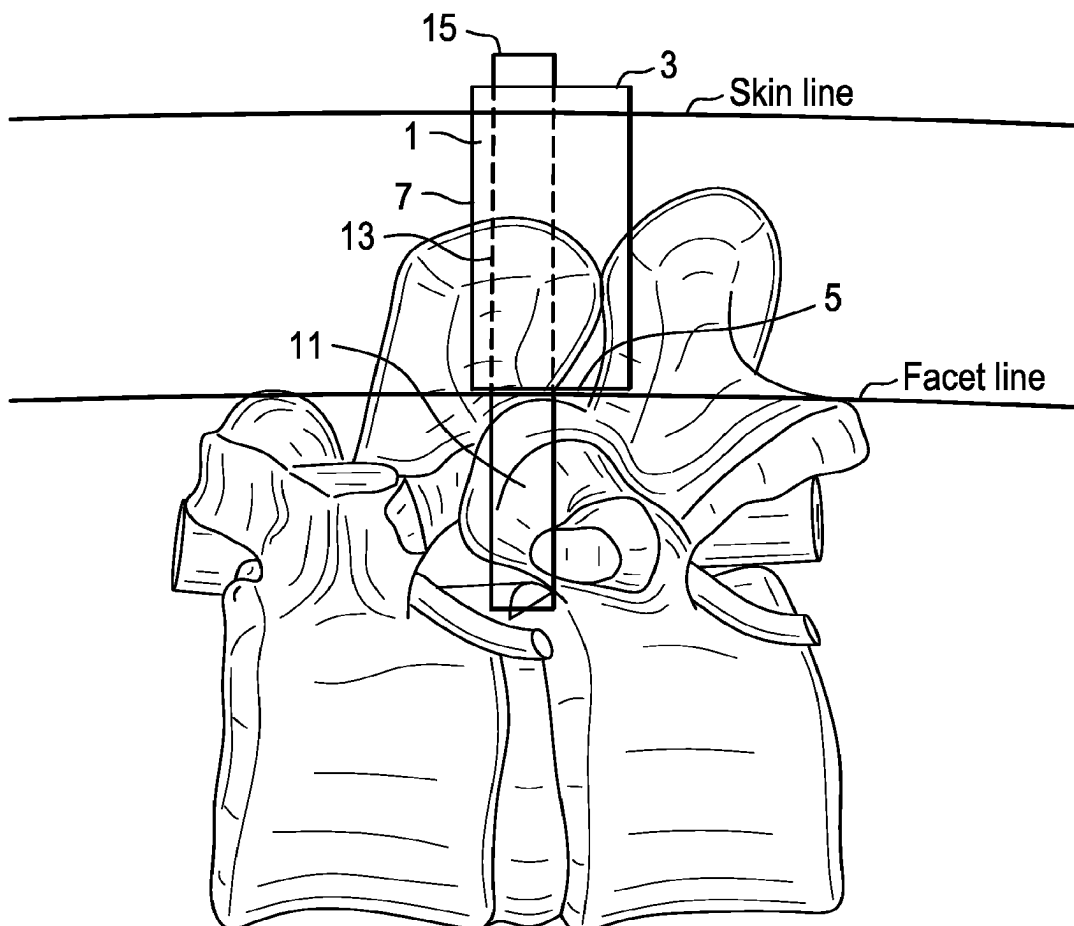


FIG. 3A

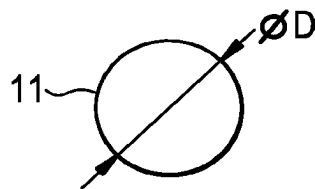


FIG. 3B

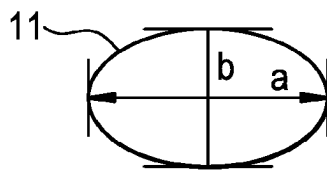


FIG. 3C

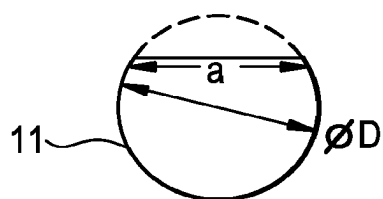


FIG. 3D

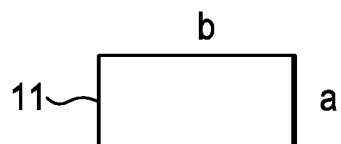


FIG. 4

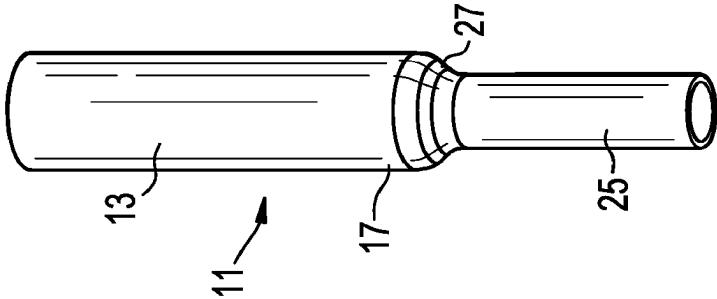


FIG. 5

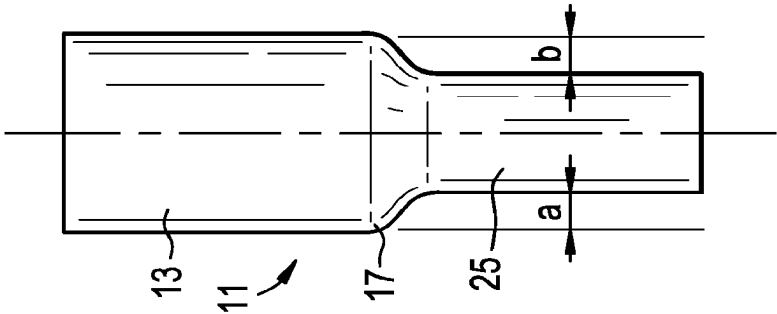


FIG. 6

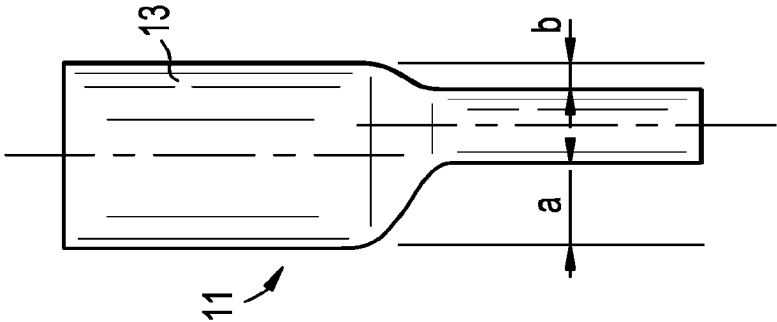


FIG. 7

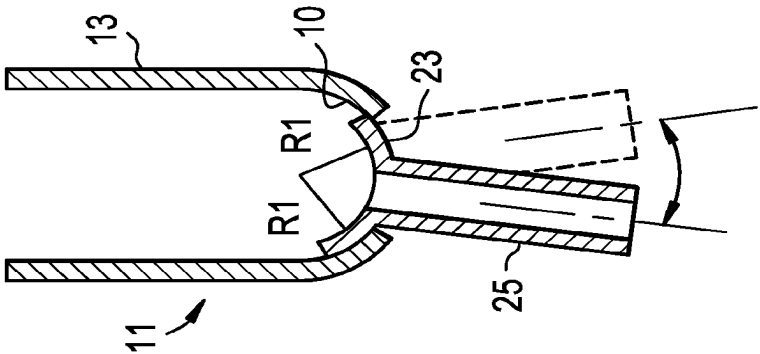


FIG. 9

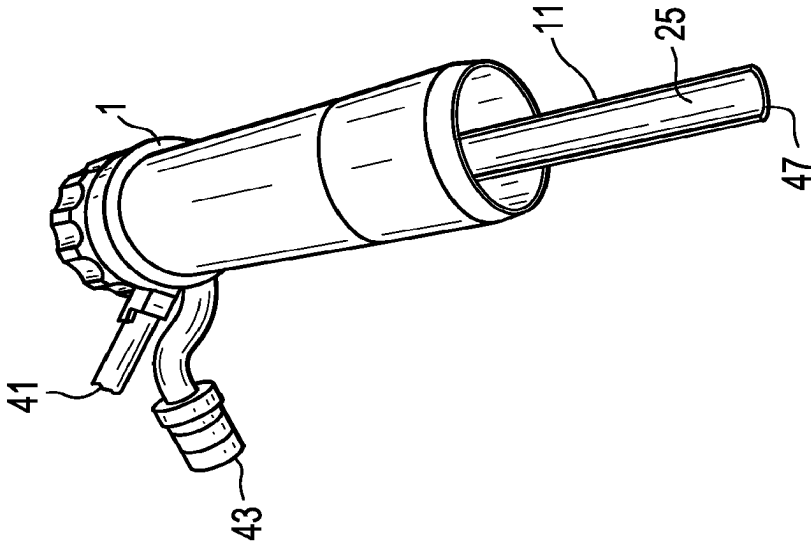


FIG. 8B

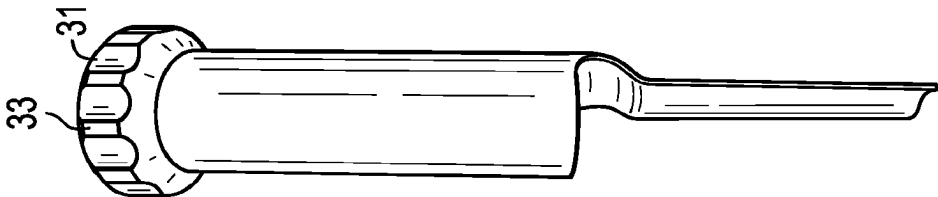


FIG. 8A

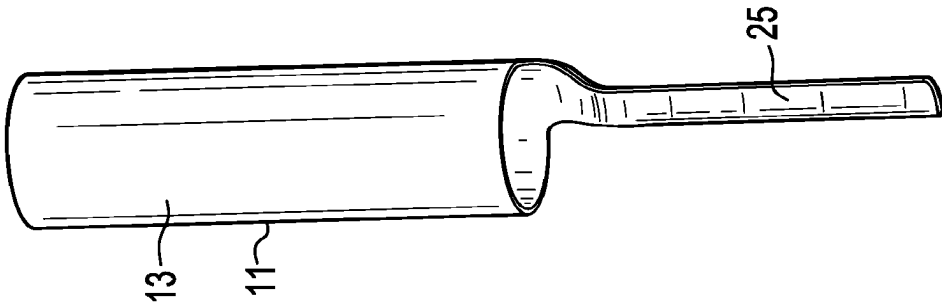


FIG. 10

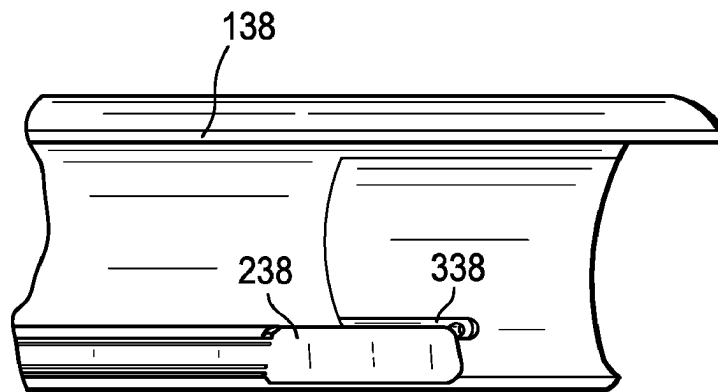


FIG. 11

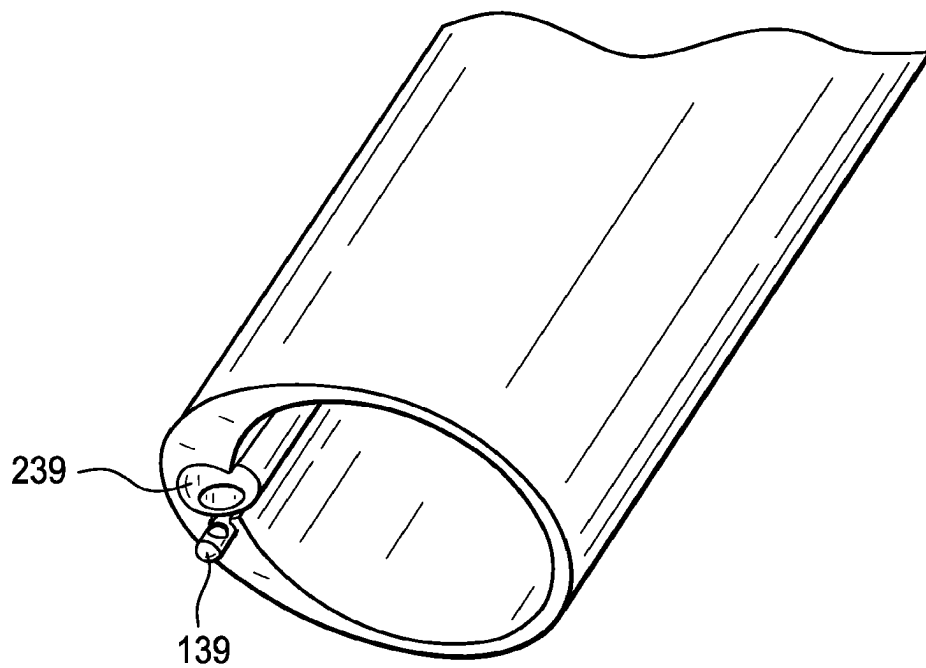


FIG. 12

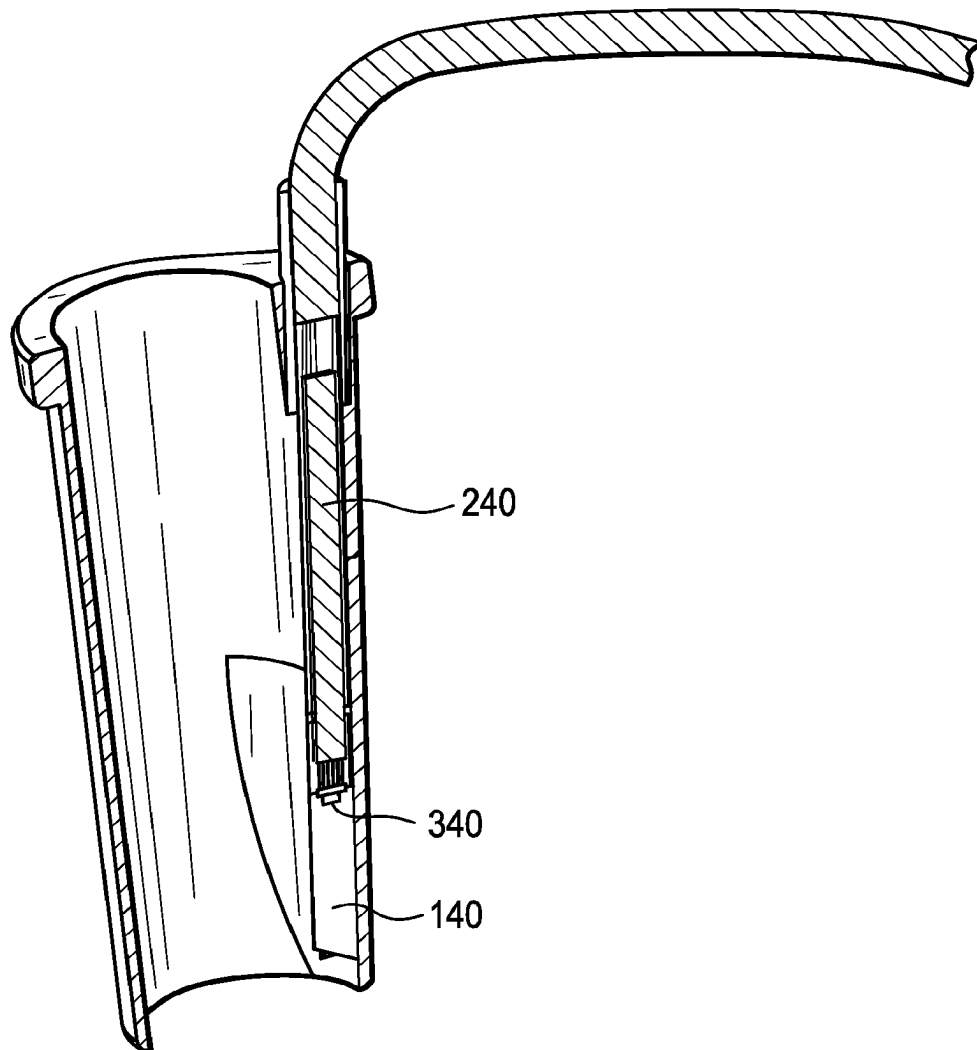
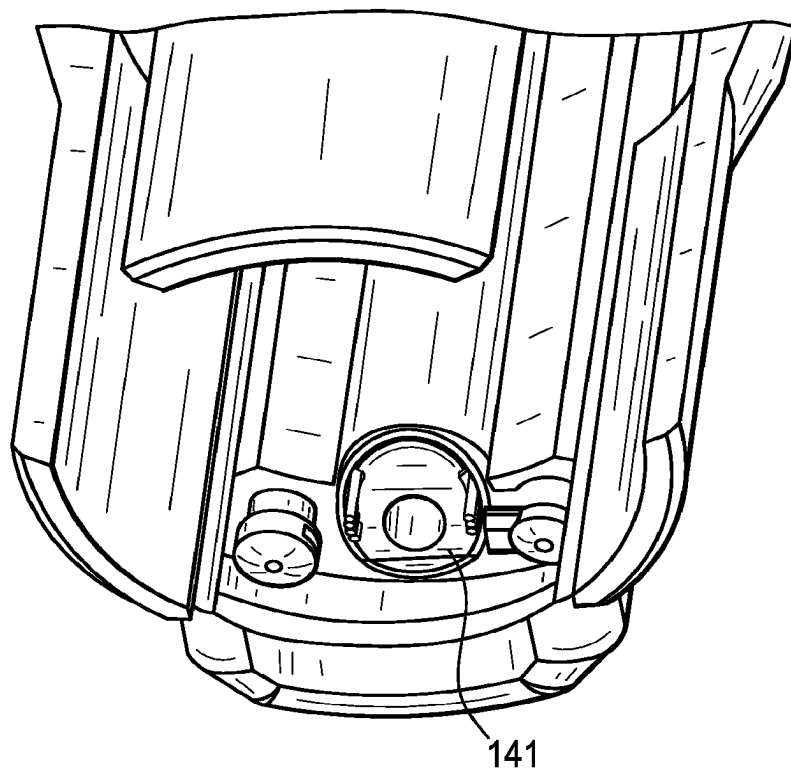


FIG. 13



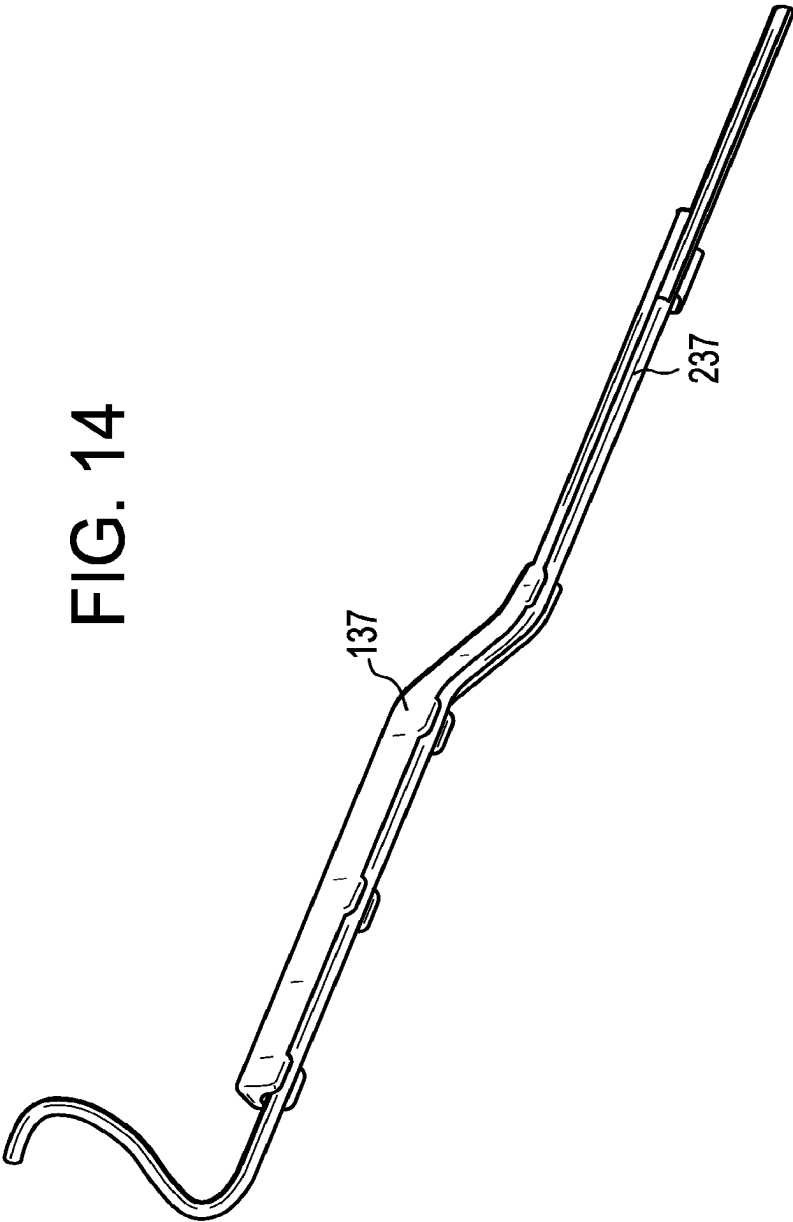


FIG. 16

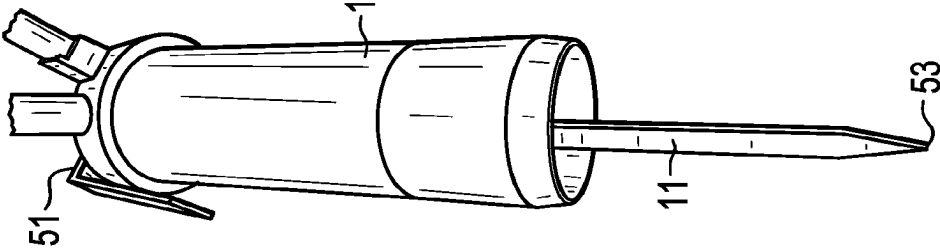


FIG. 15B



FIG. 15A

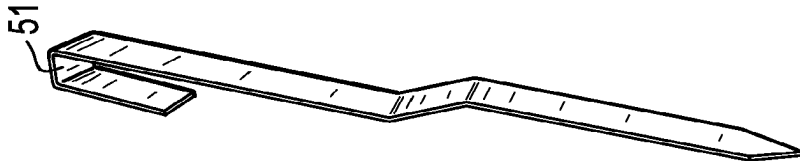


FIG. 17

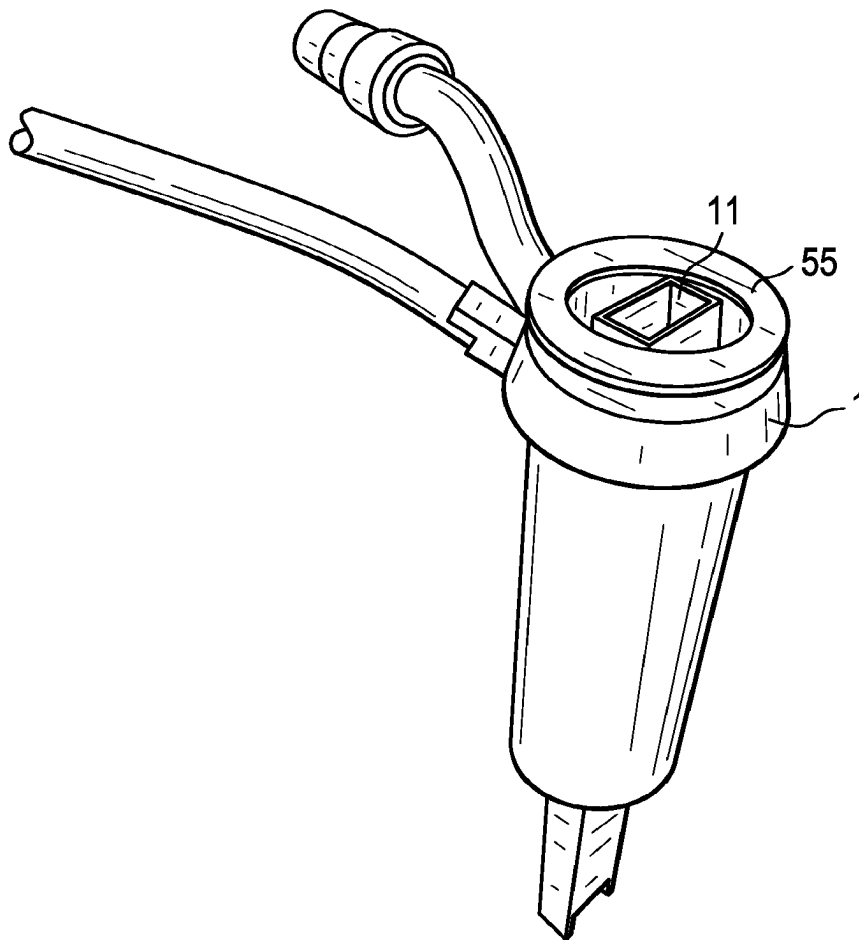


FIG. 18

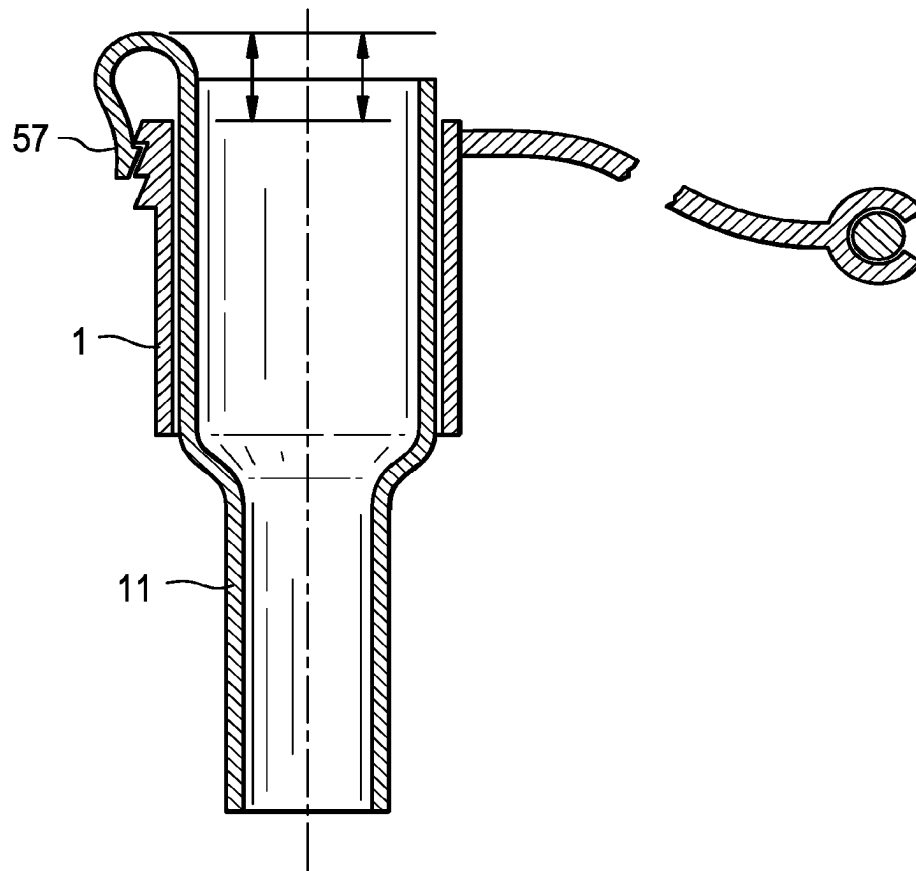


FIG. 19

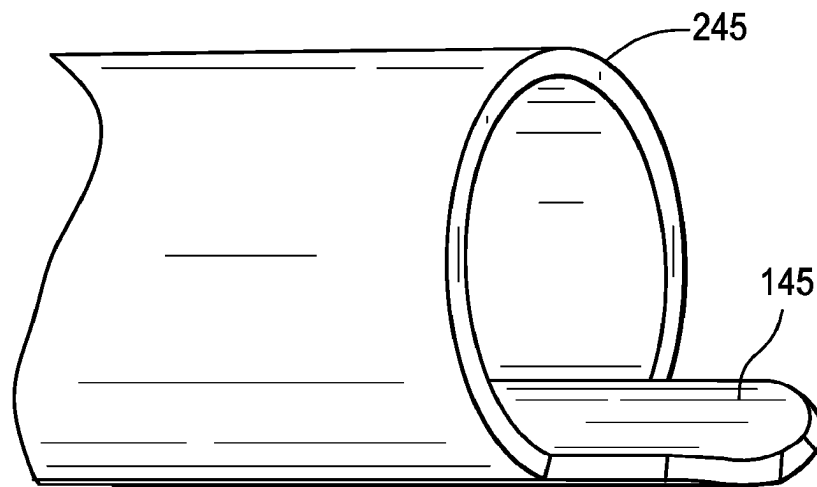


FIG. 20

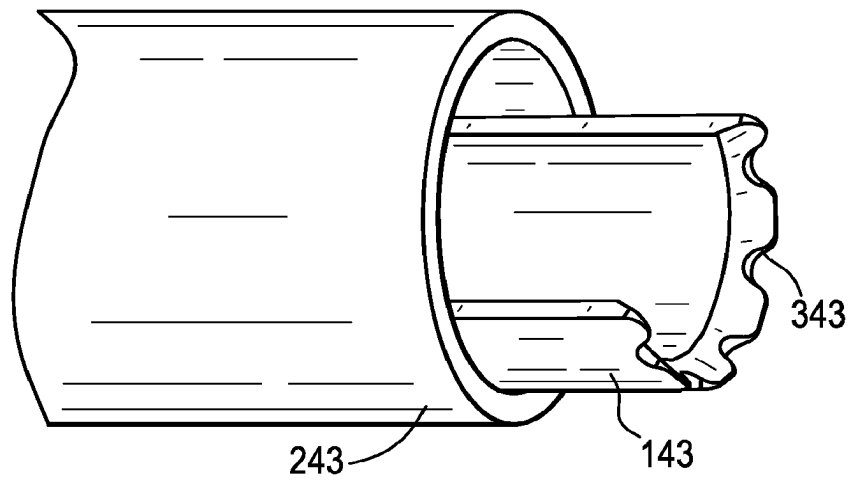


FIG. 21

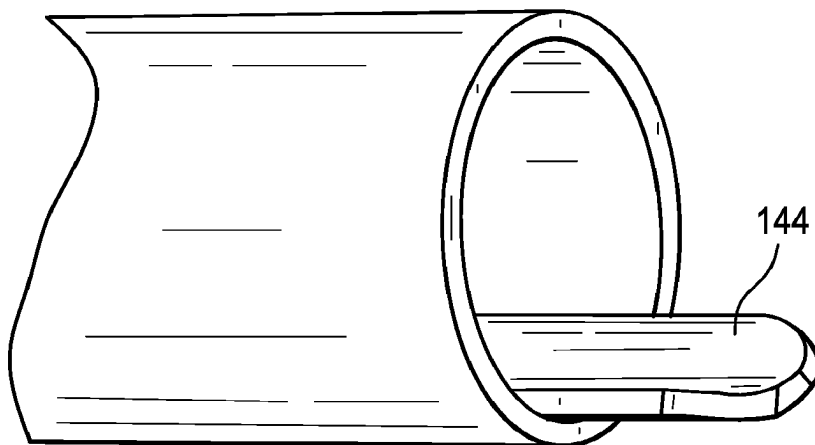


FIG. 22

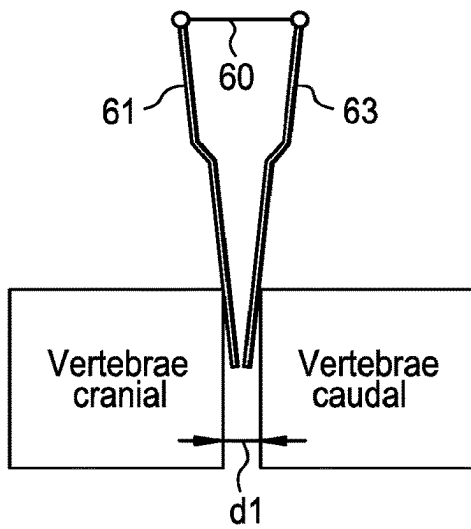


FIG. 23

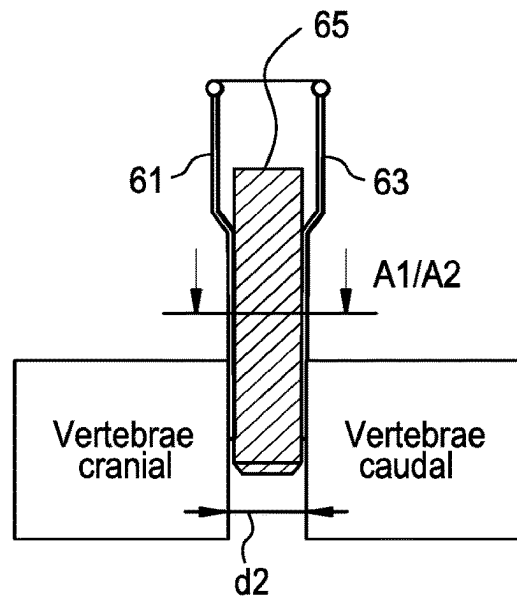


FIG. 24A

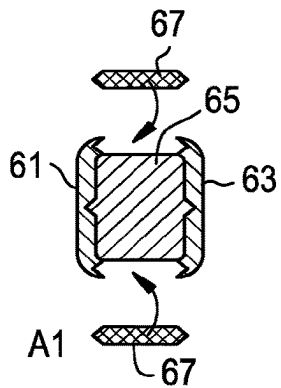


FIG. 24B

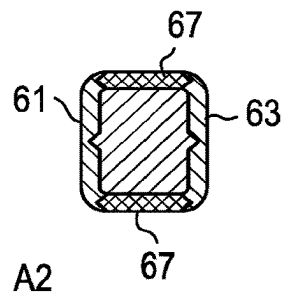


FIG. 26

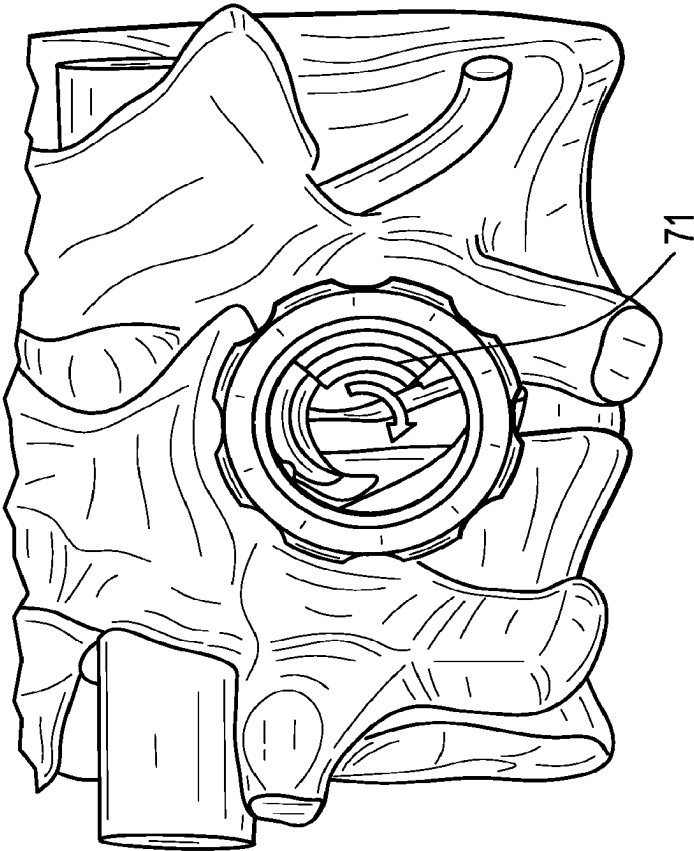


FIG. 25

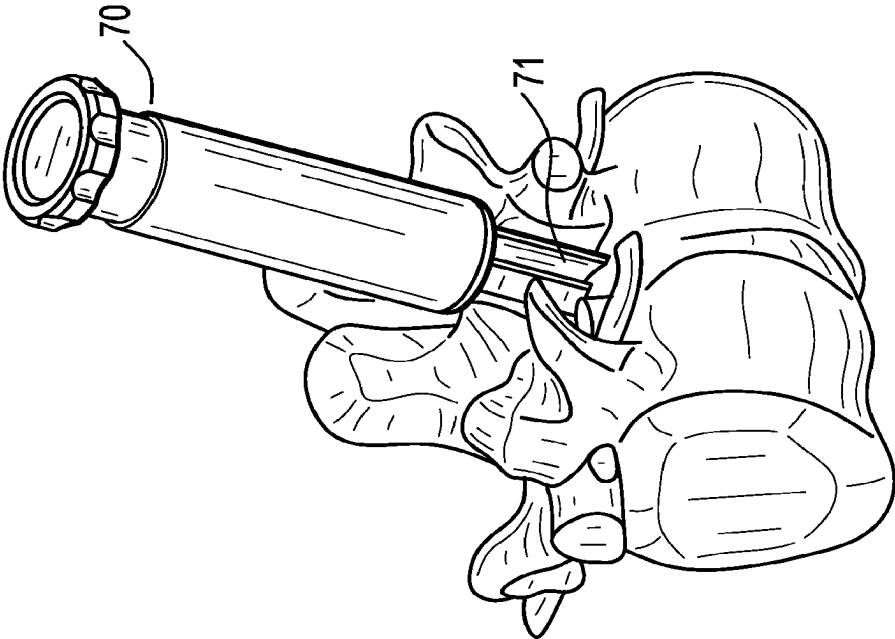


FIG. 27

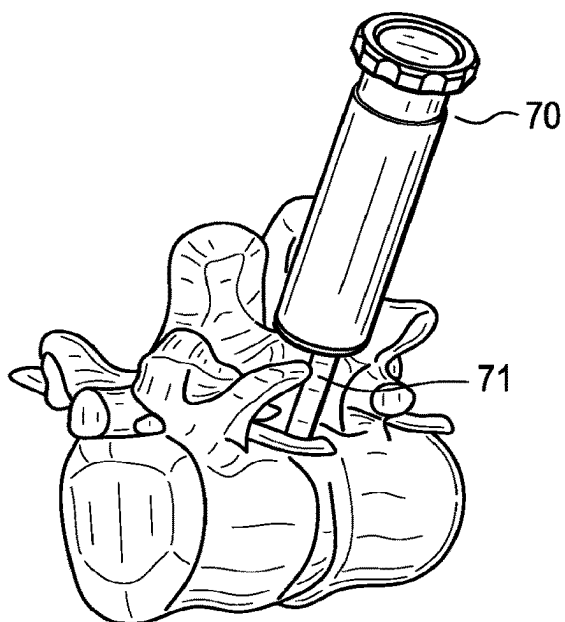


FIG. 29

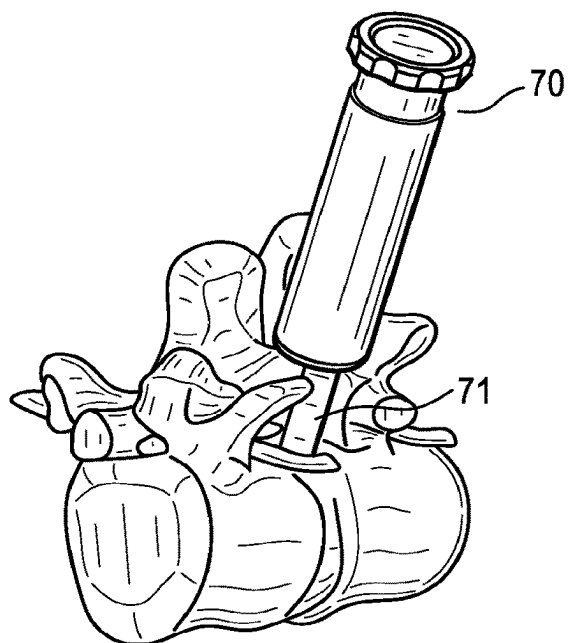


FIG. 28

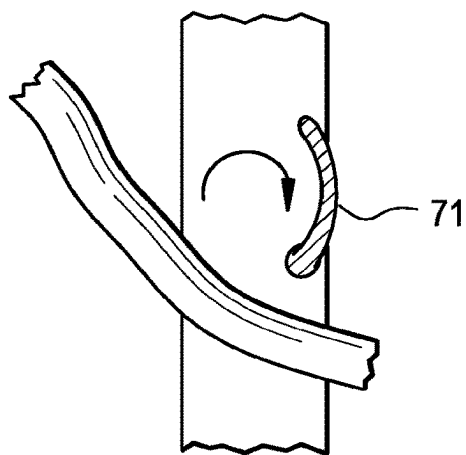


FIG. 30

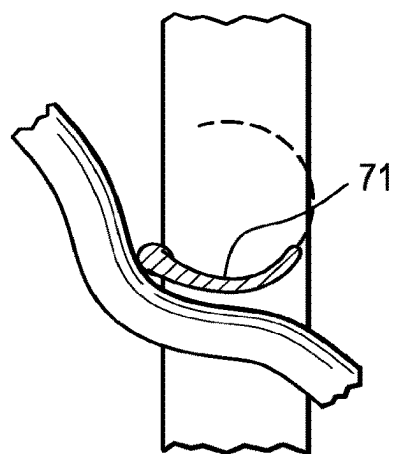


FIG. 31

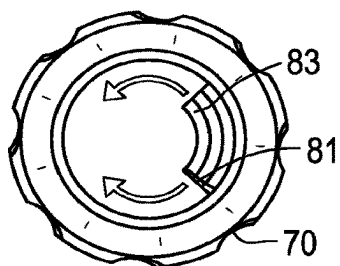


FIG. 33

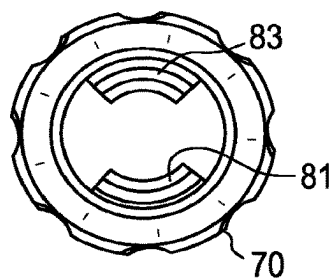


FIG. 32

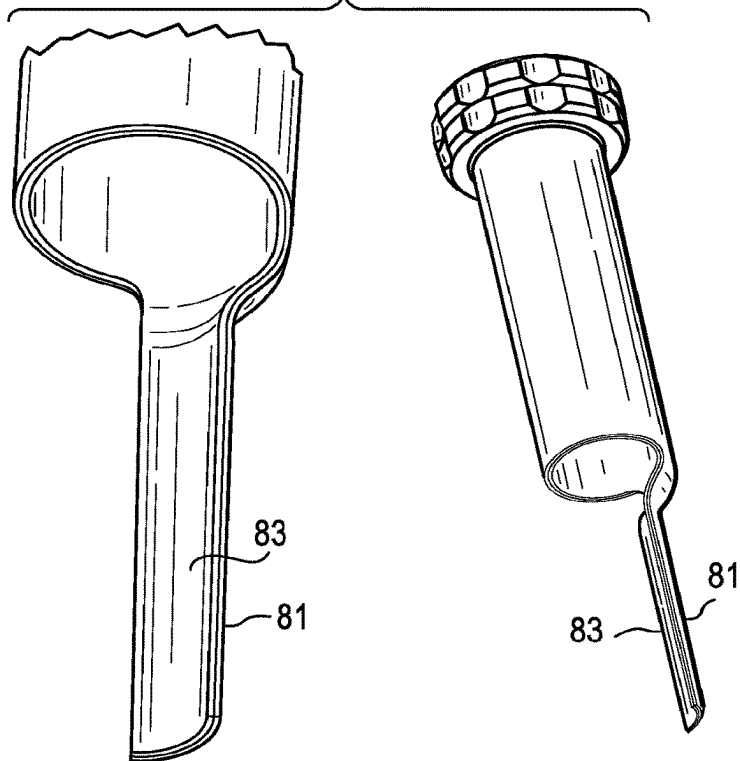


FIG. 34

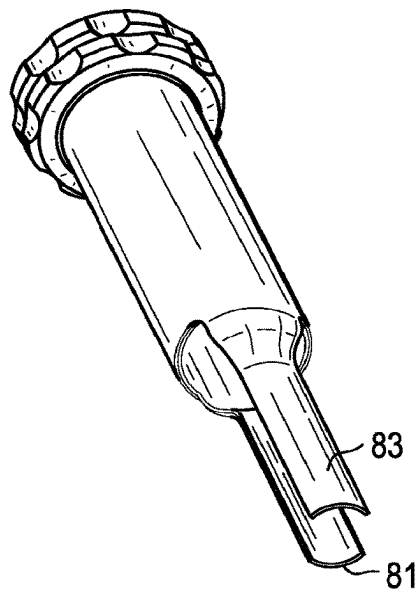


FIG. 35

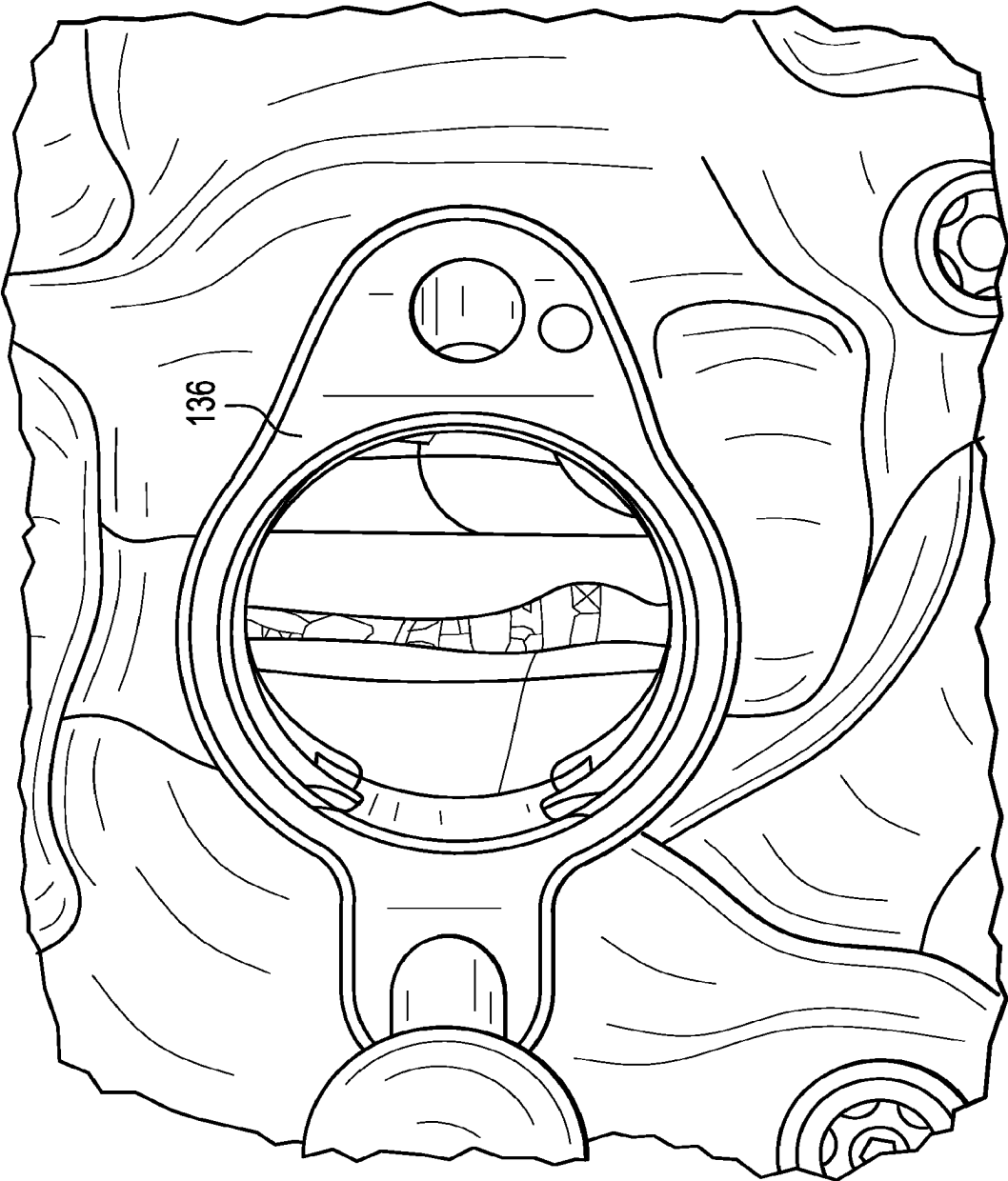


FIG. 36

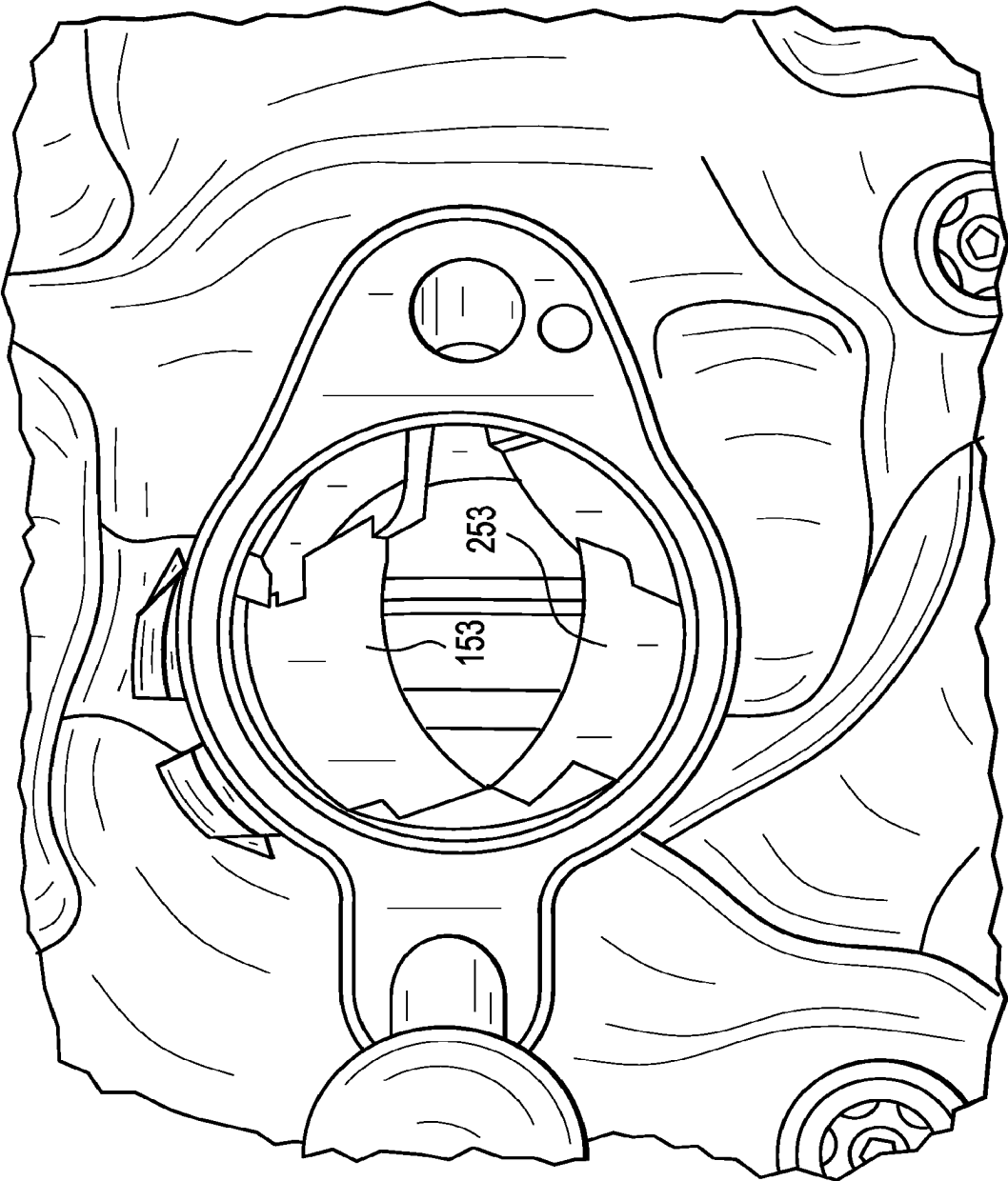


FIG. 37

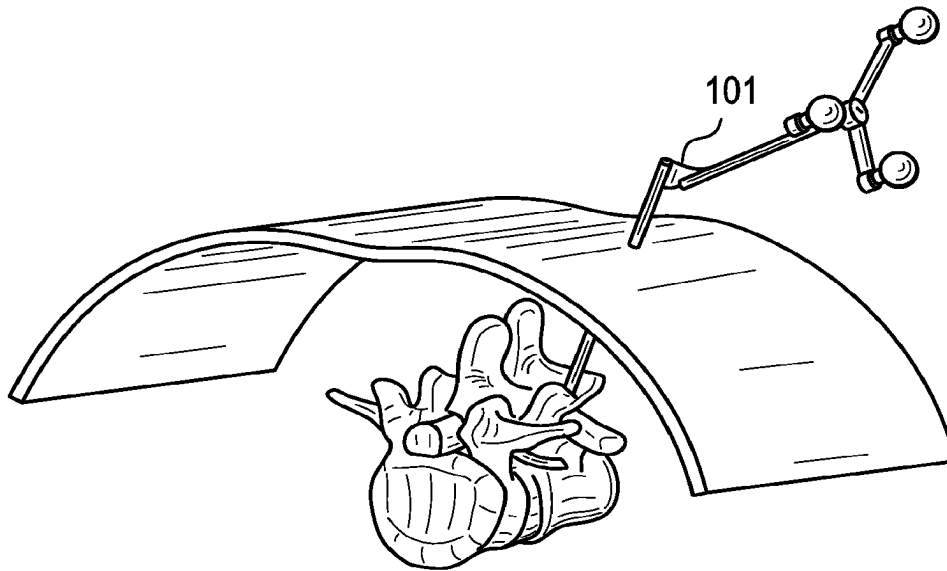


FIG. 38

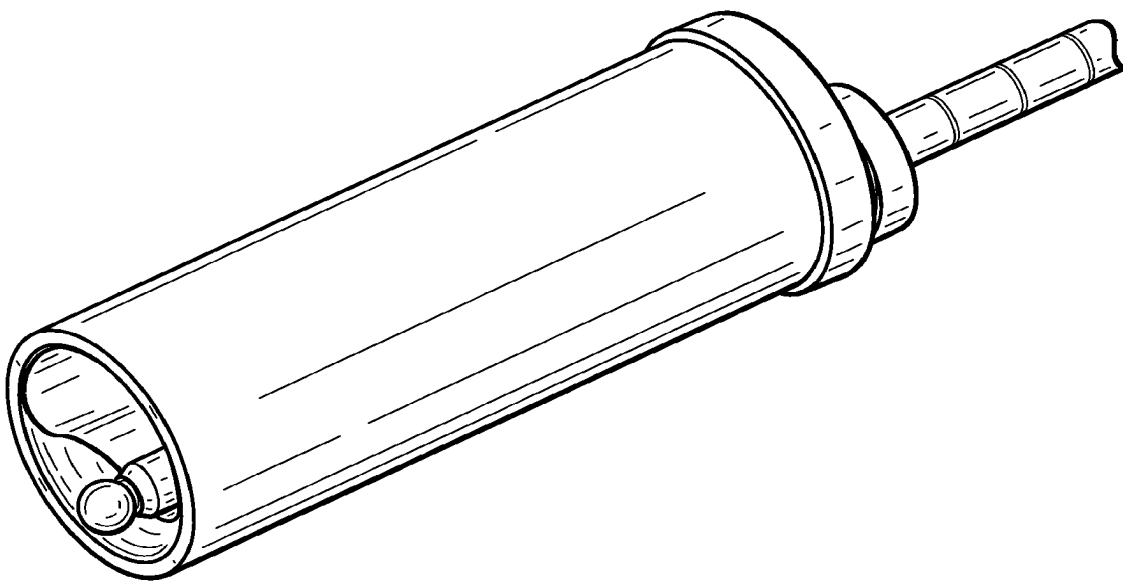


FIG. 39A

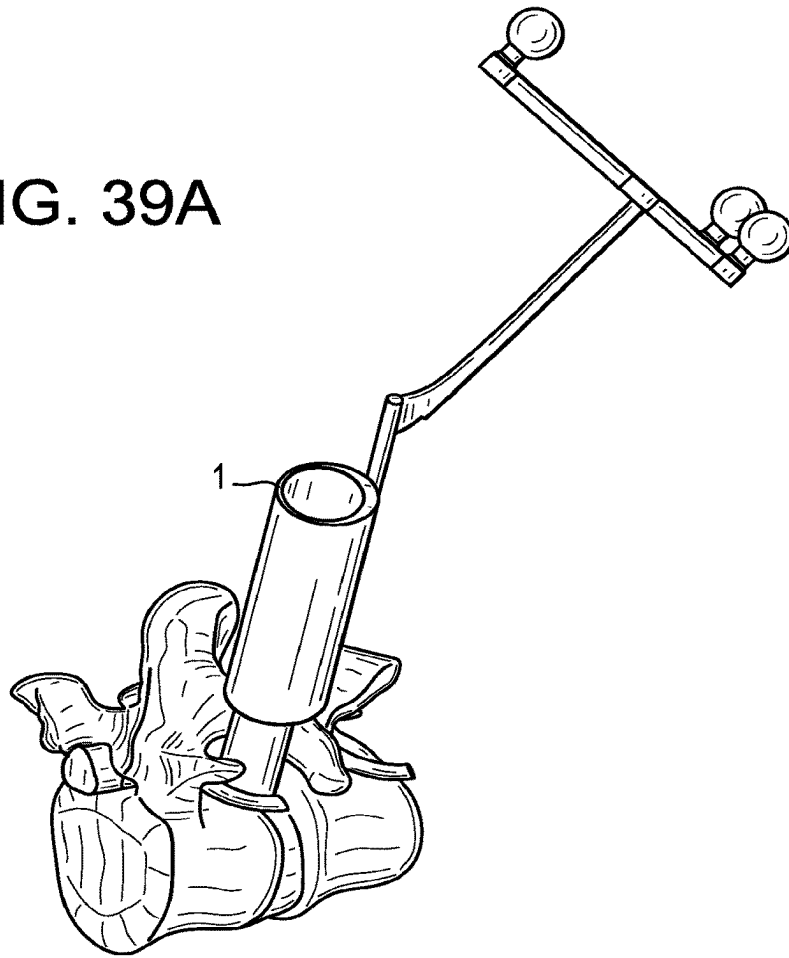


FIG. 39B

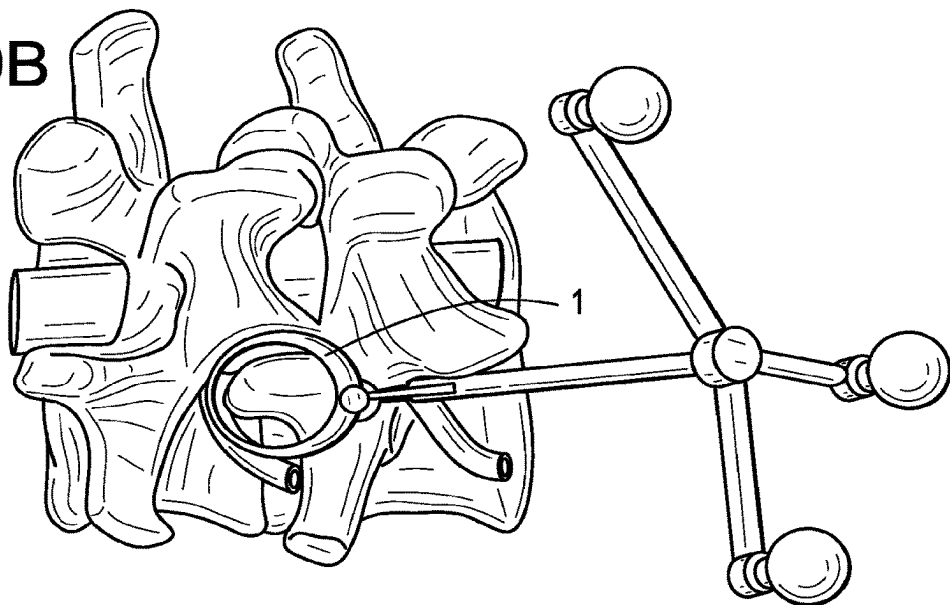


FIG. 40

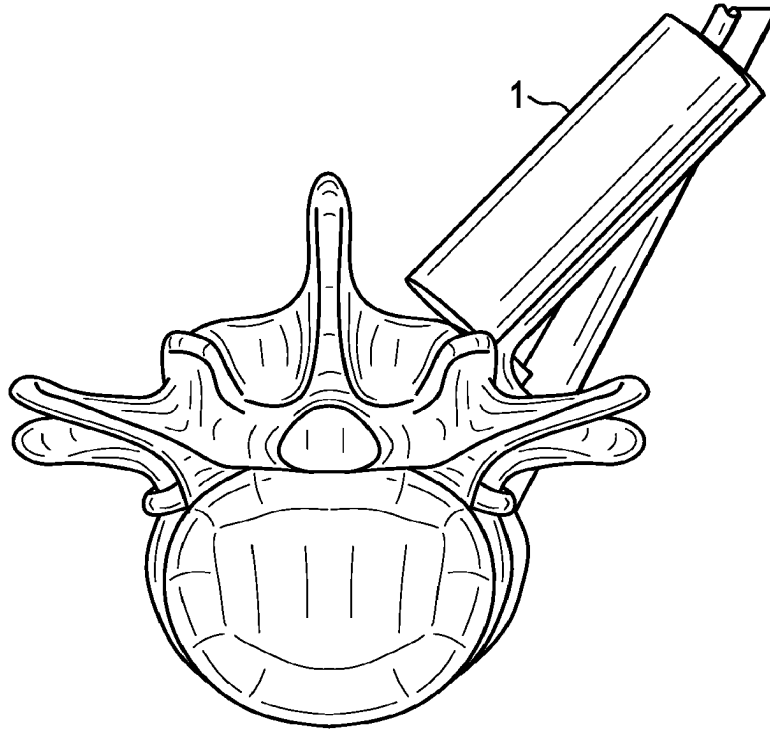


FIG. 41

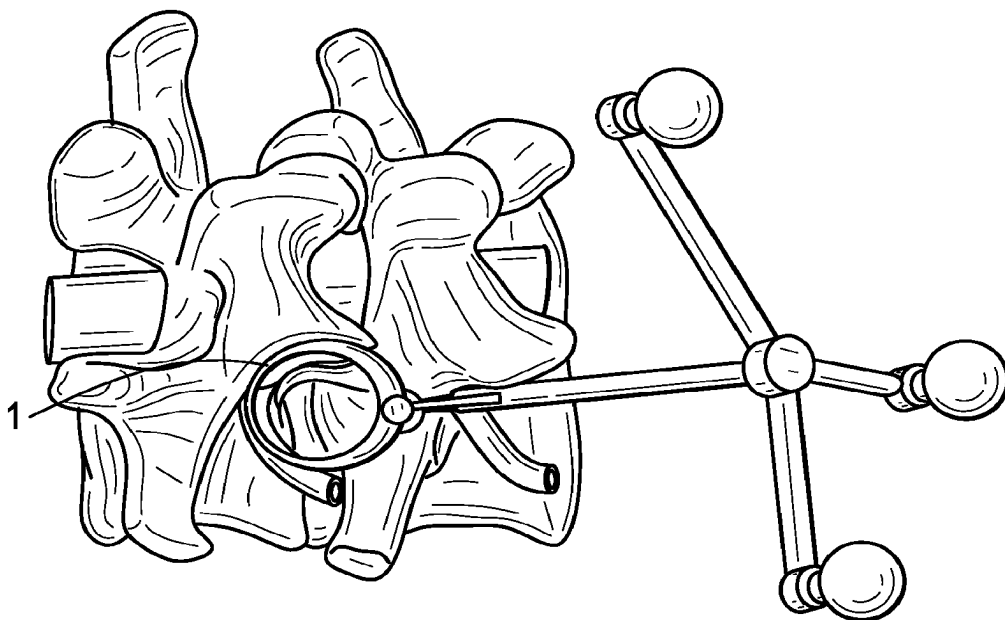


FIG. 42

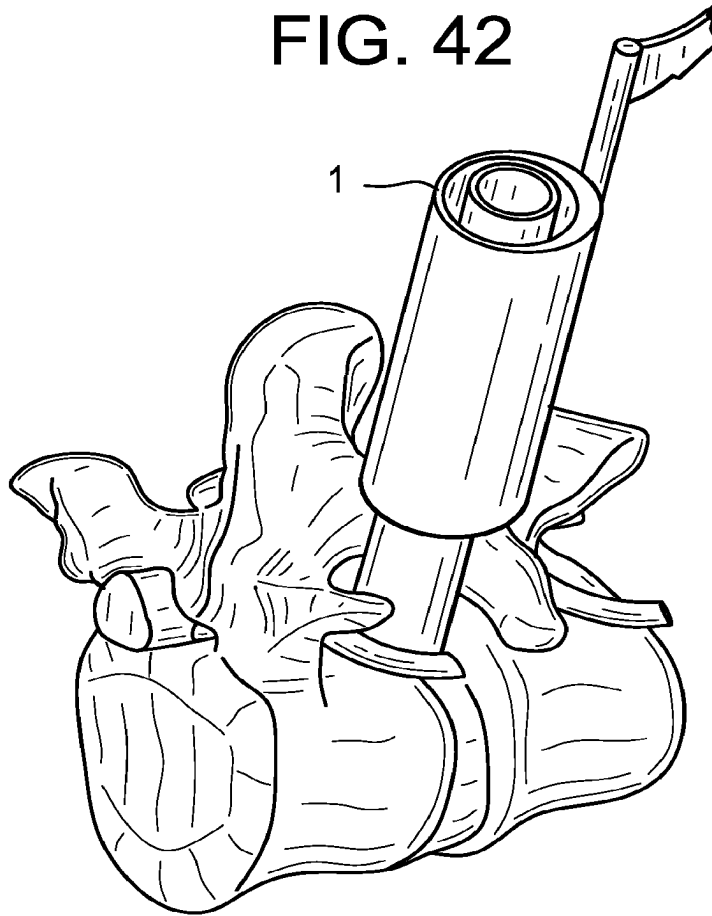


FIG. 43

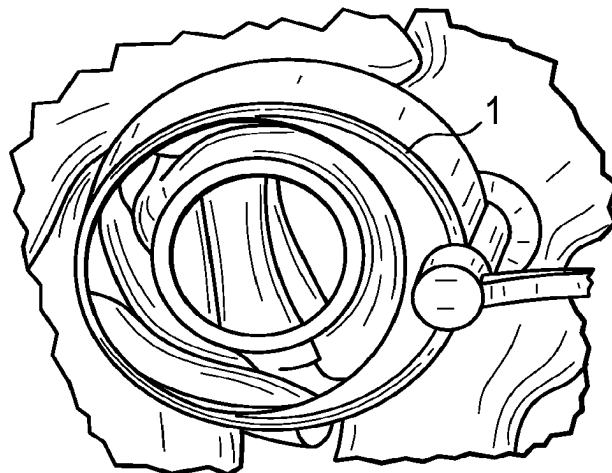


FIG. 44

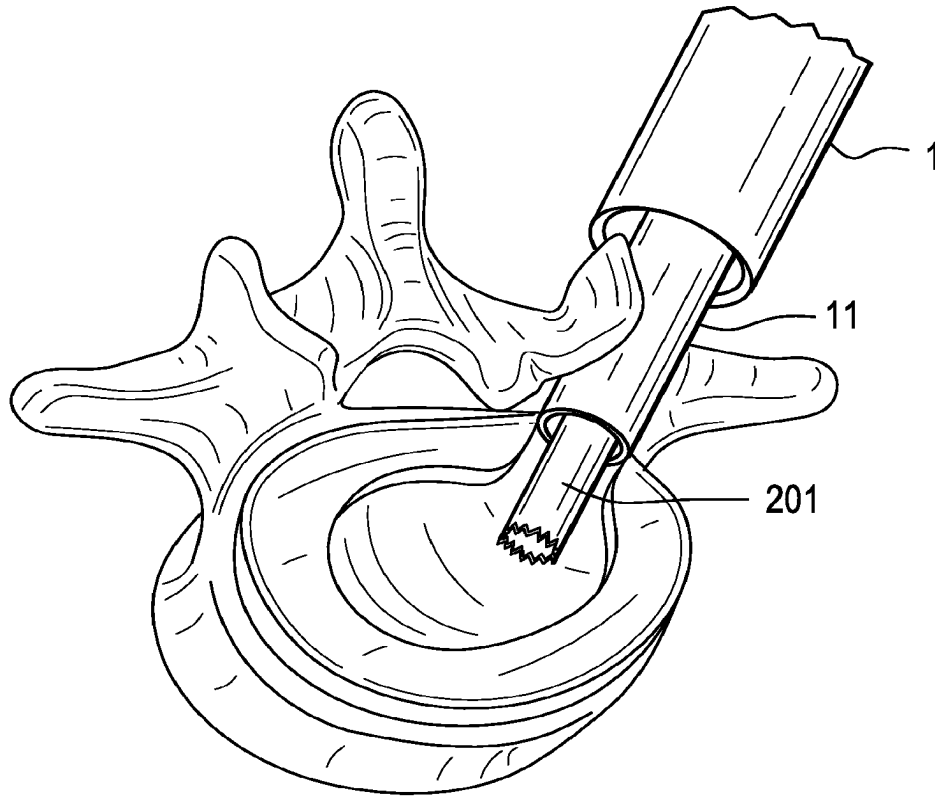


FIG. 45

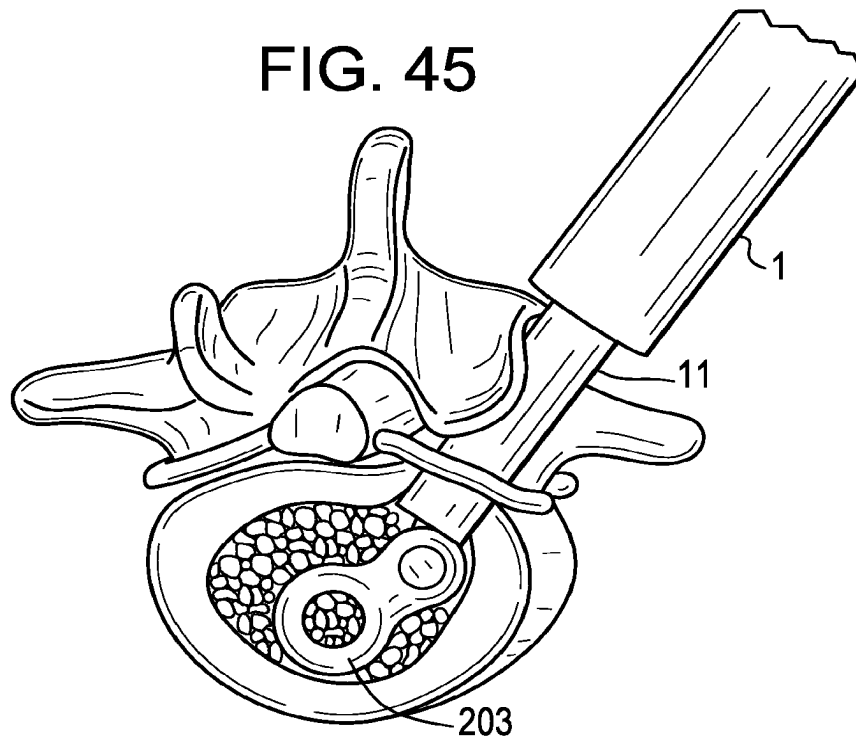
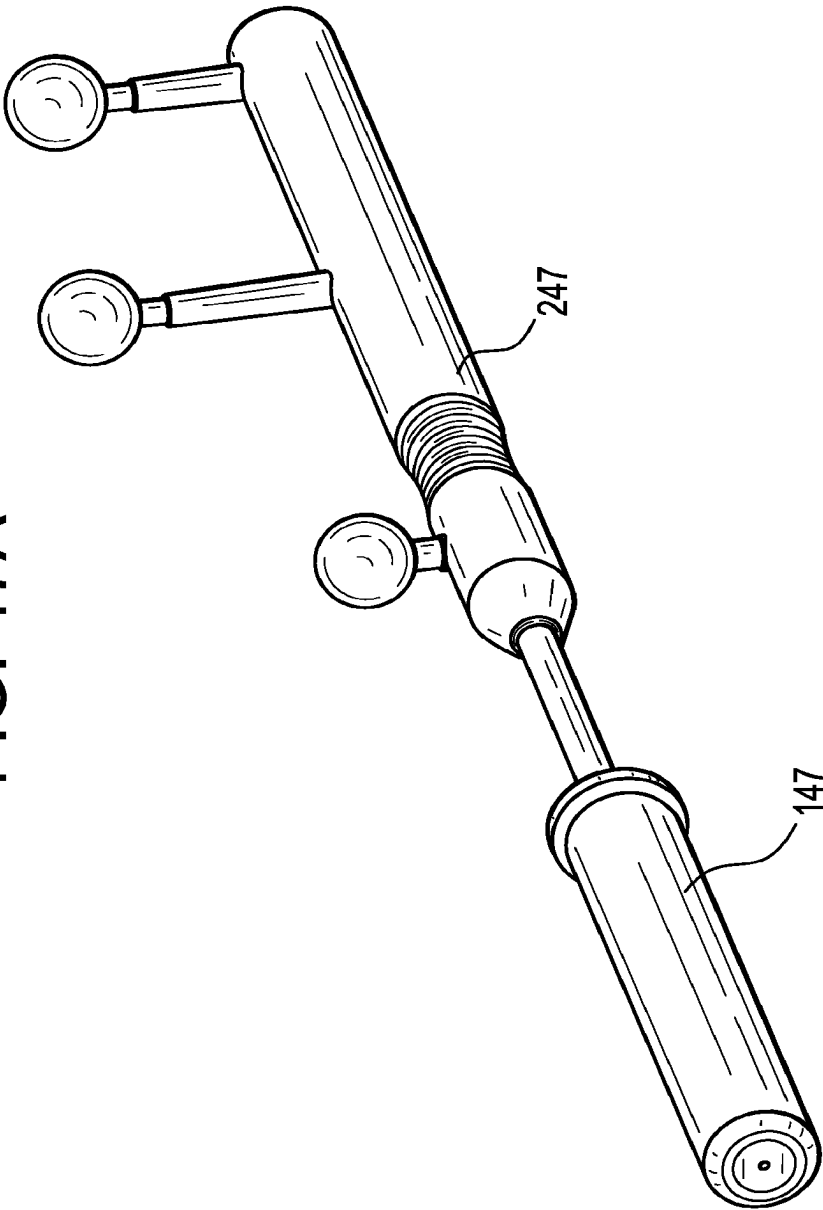
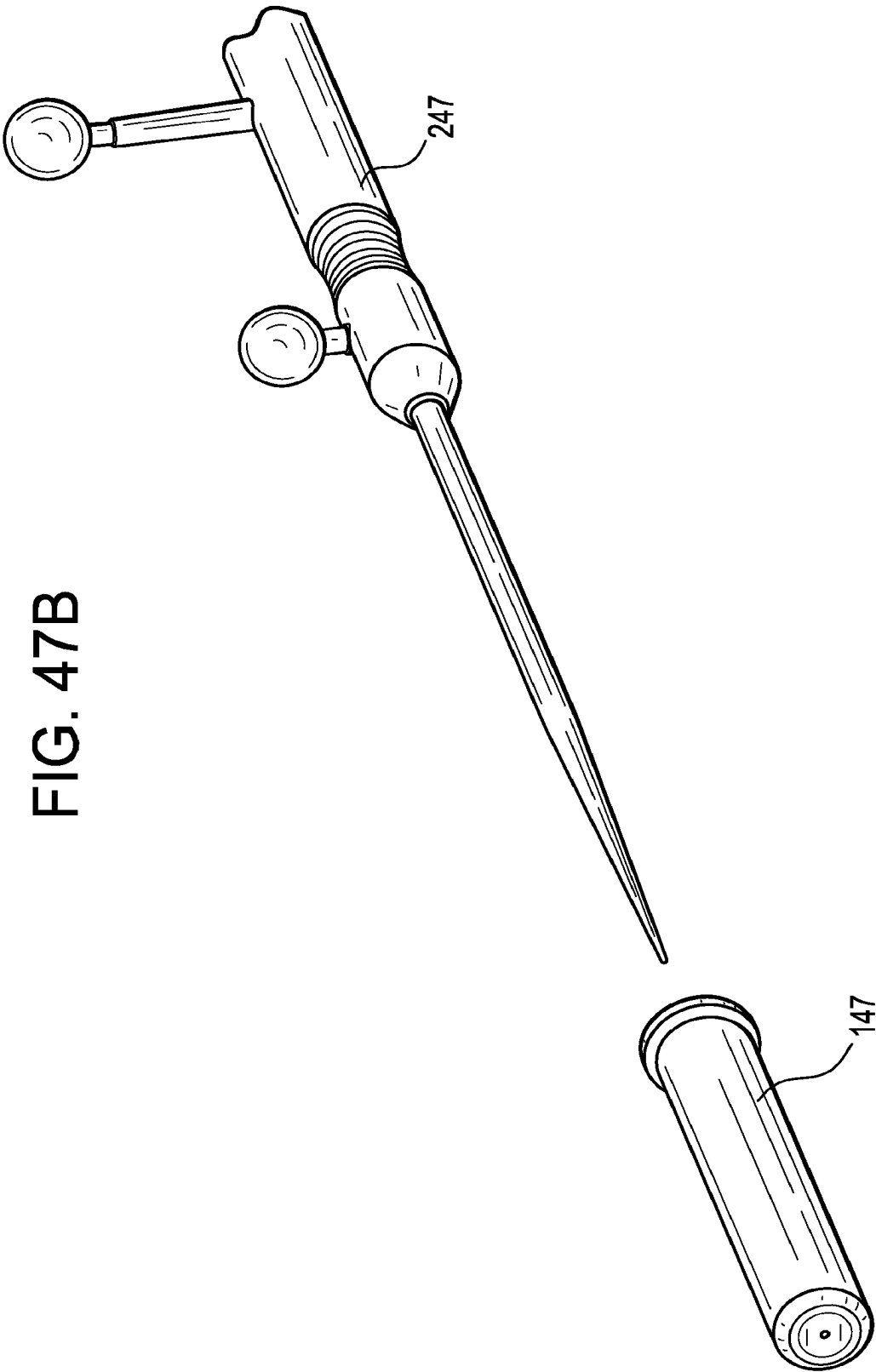


FIG. 46



FIG. 47A





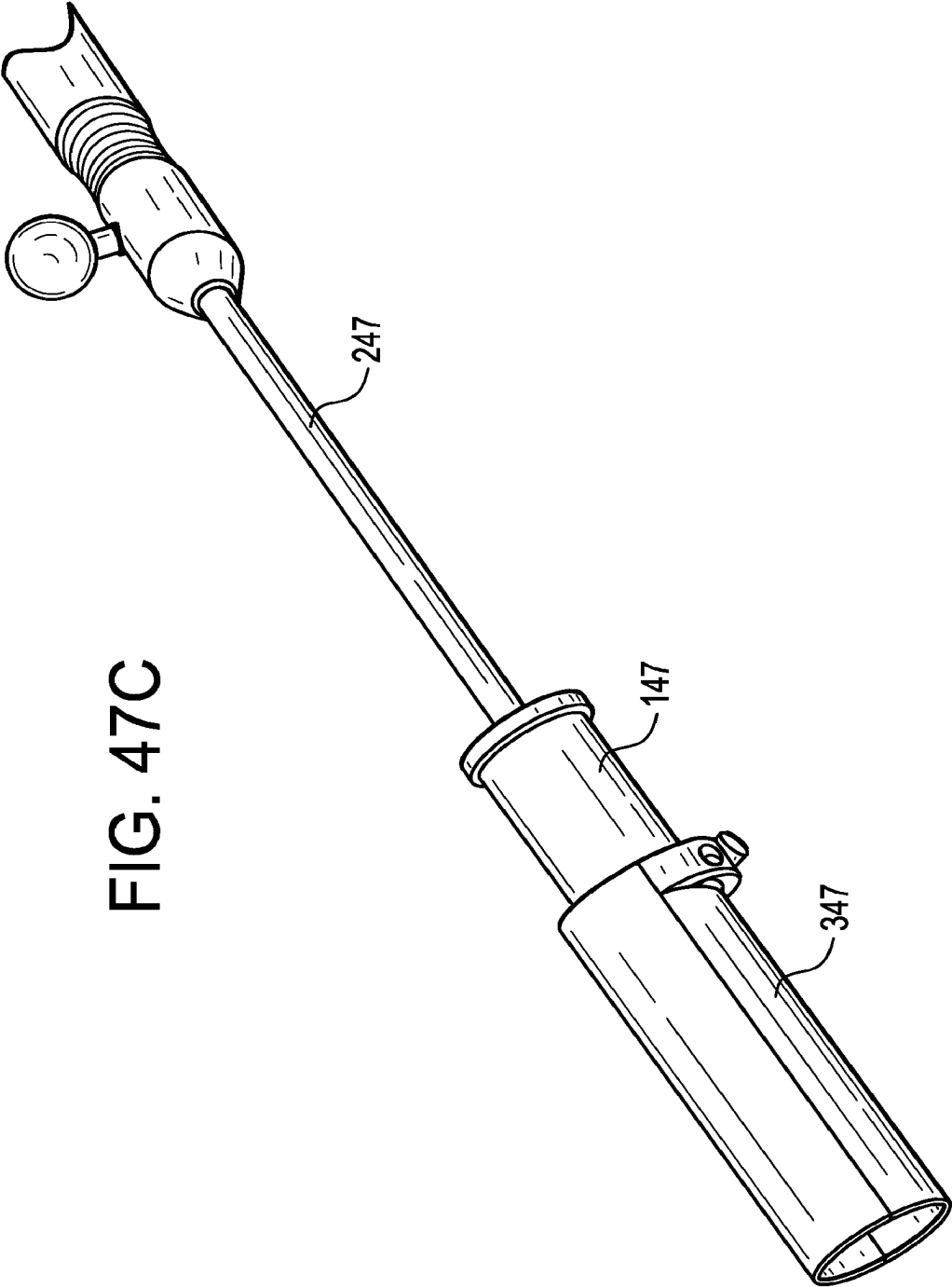


FIG. 48

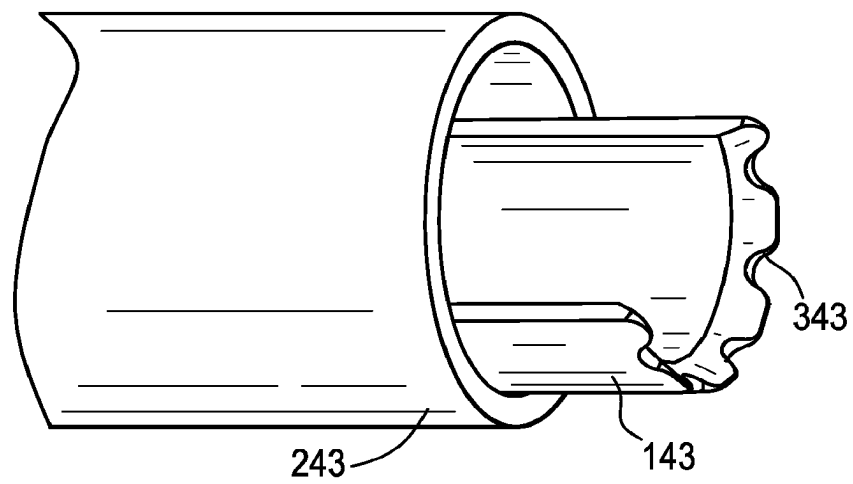


FIG. 49A

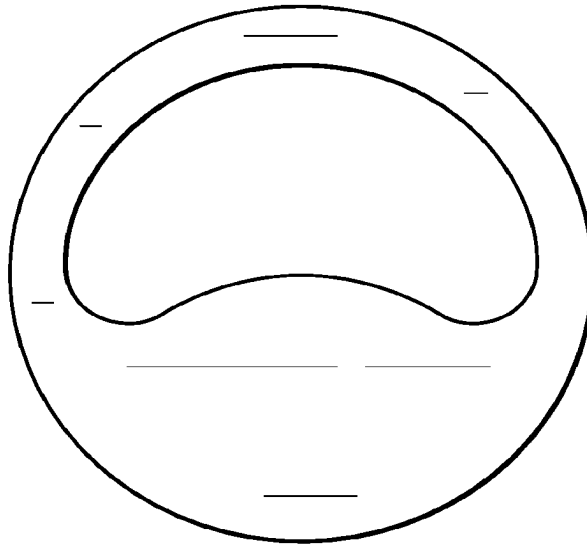


FIG. 49B

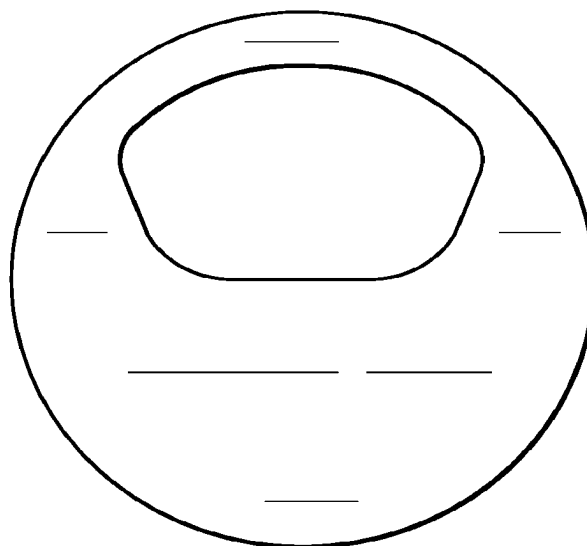
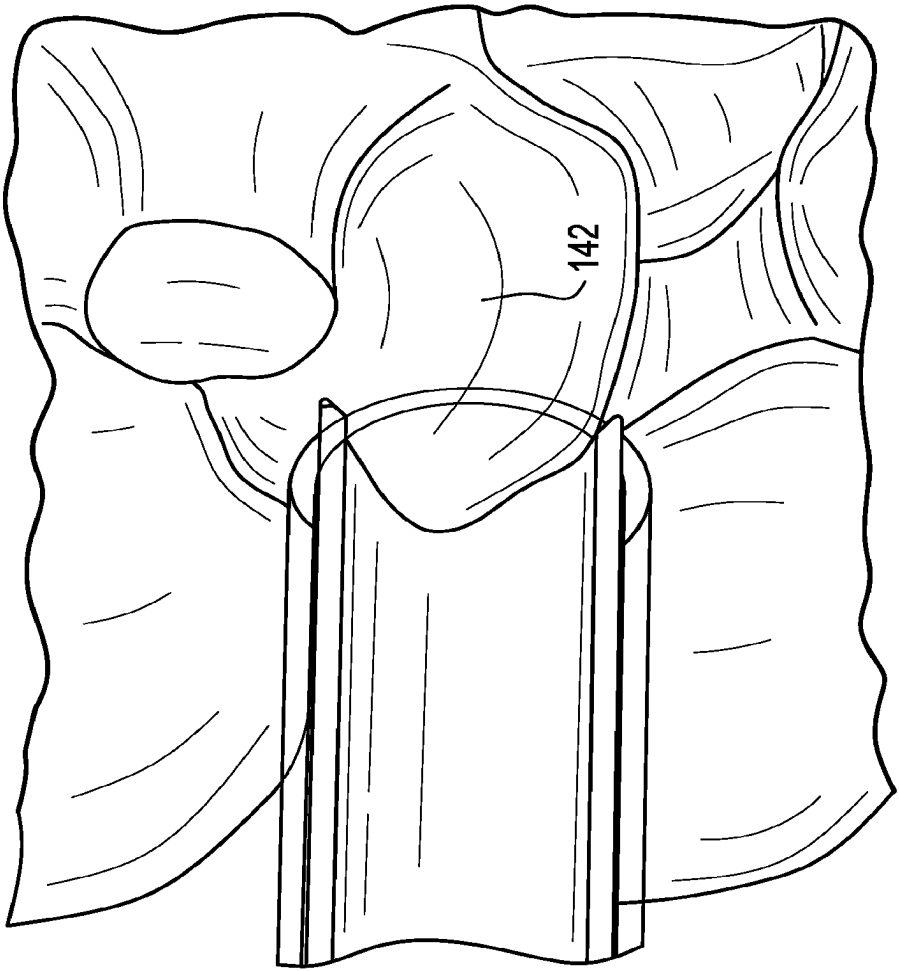


FIG. 50



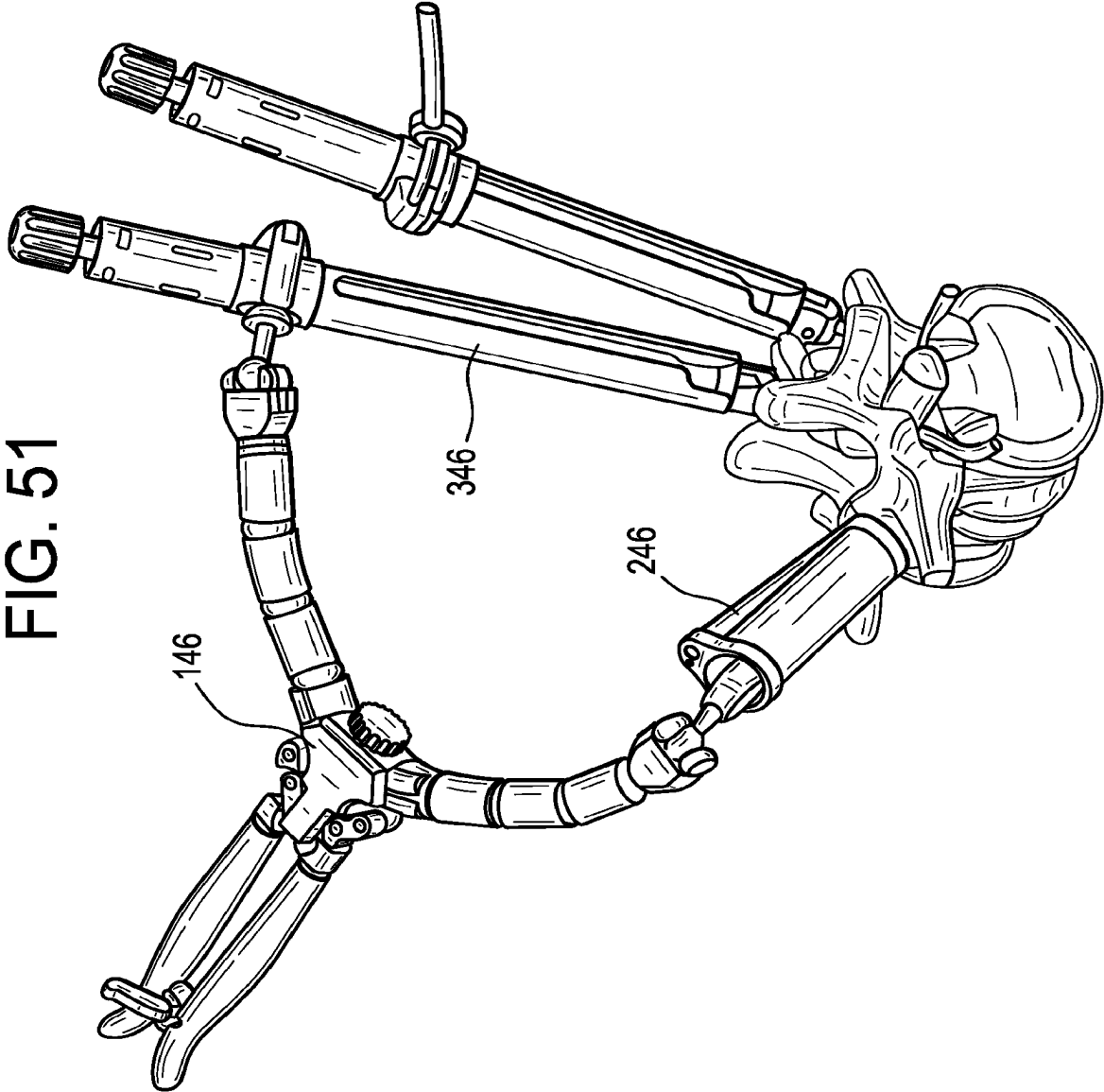


FIG. 52

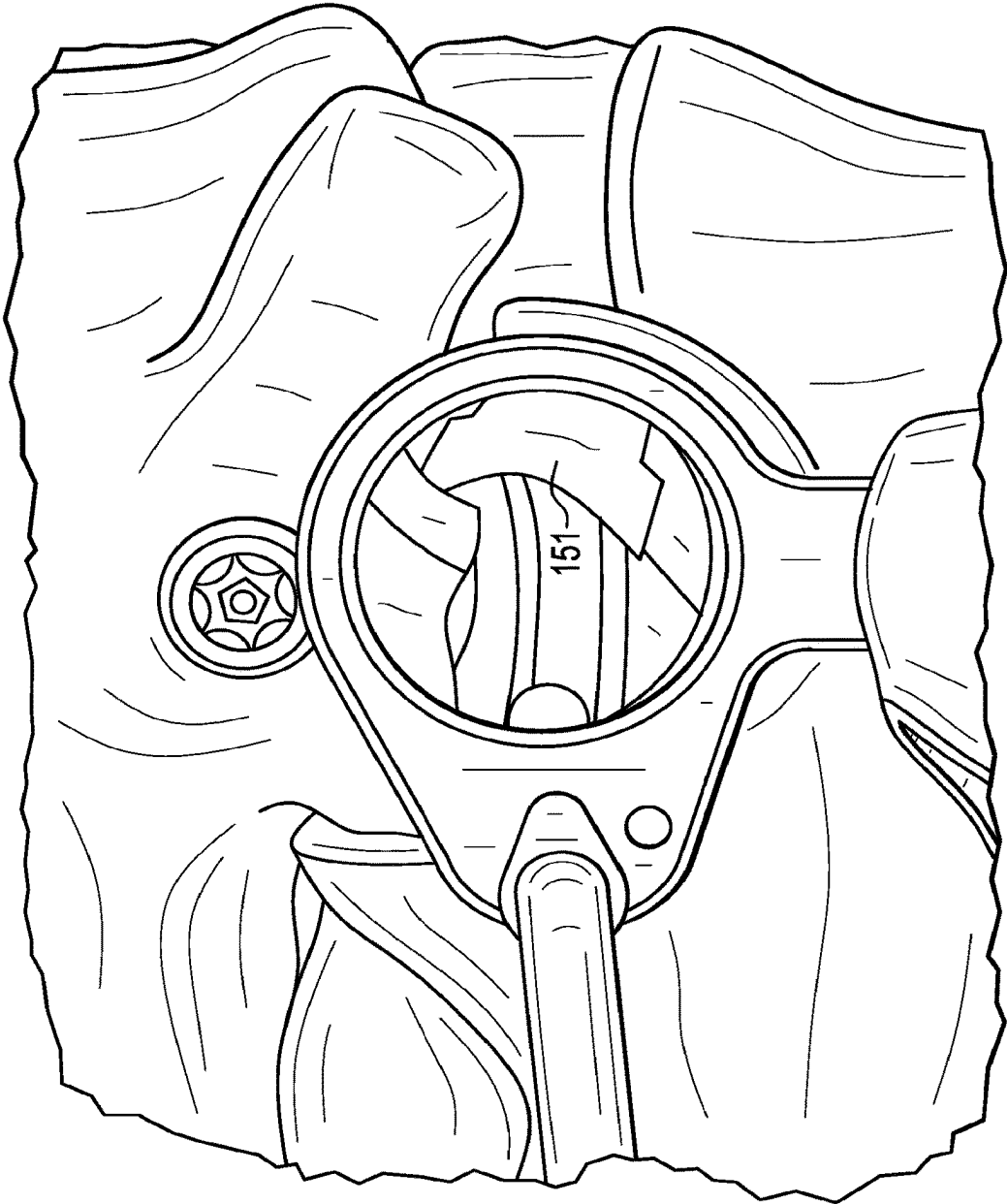


FIG. 53

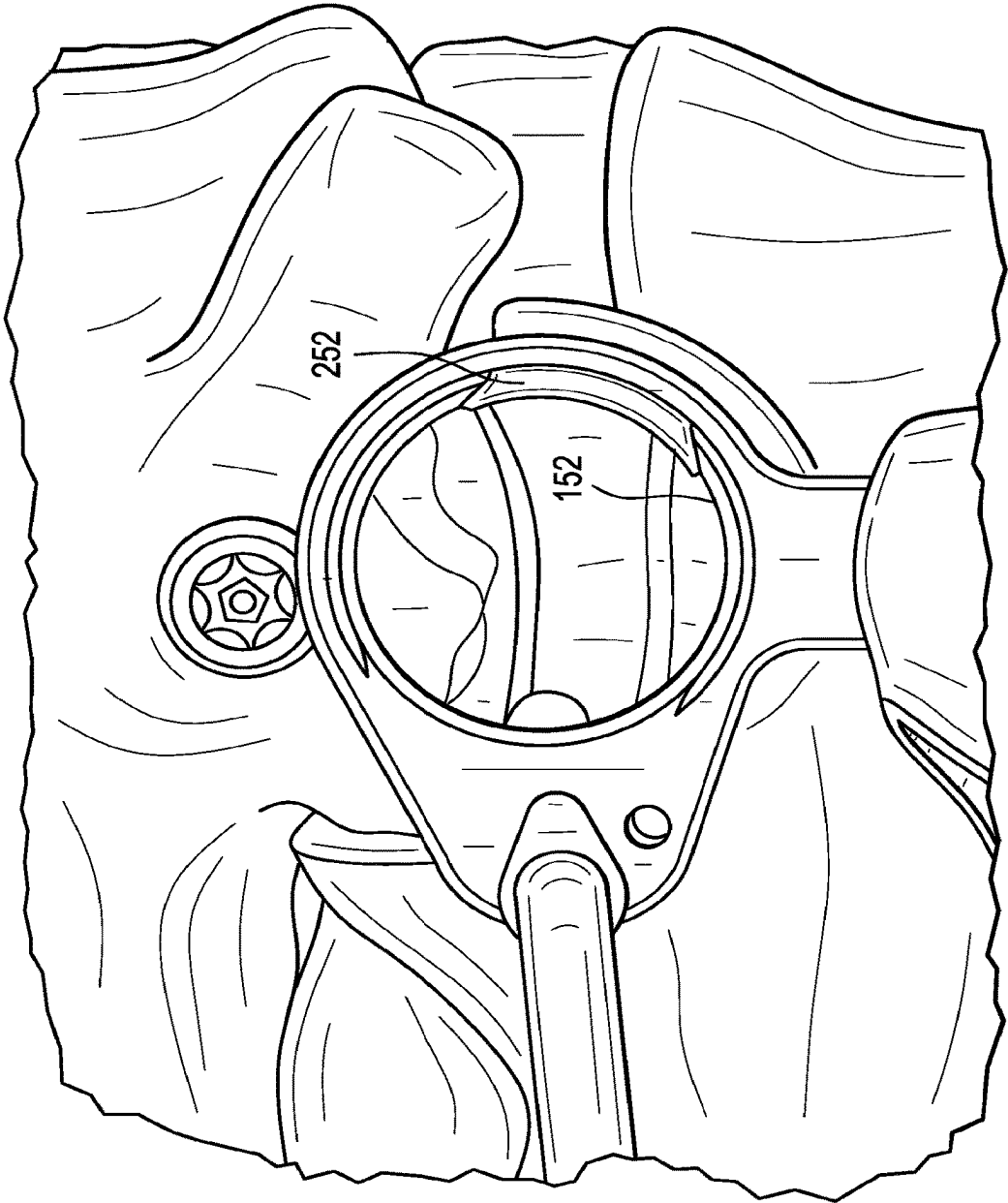
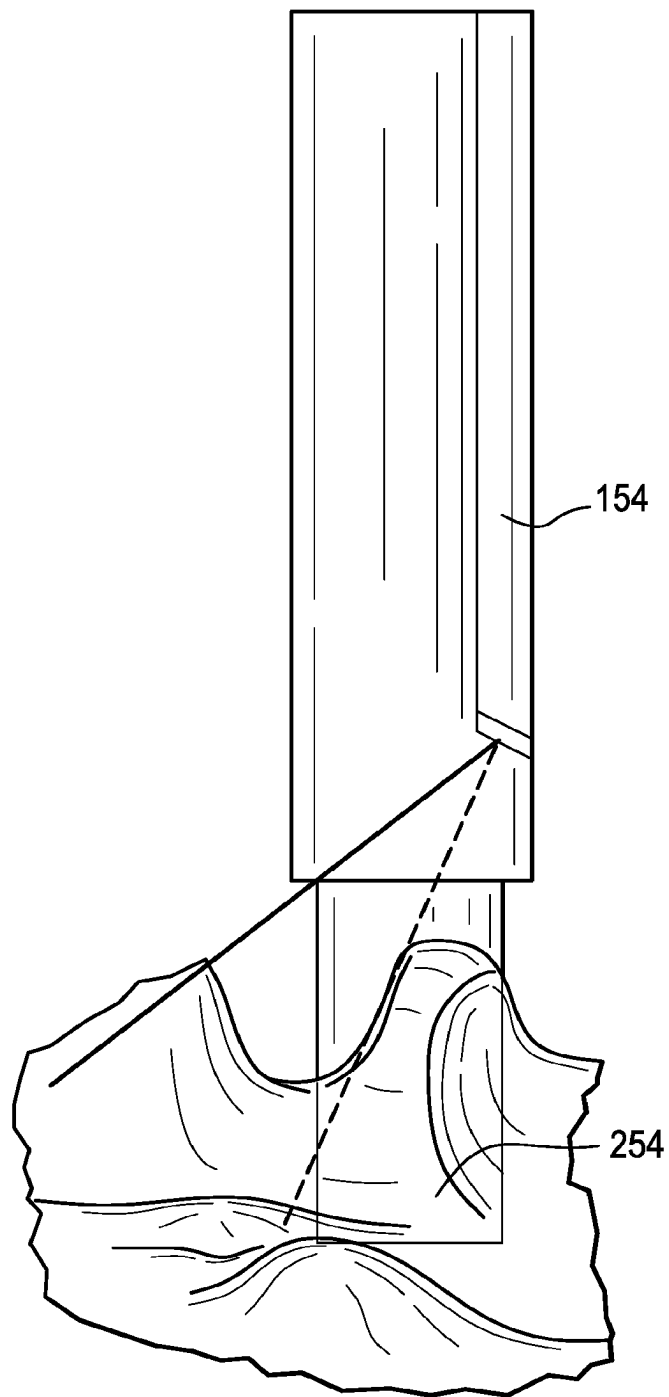


FIG. 54



MULTI-SHIELD SPINAL ACCESS SYSTEM

CONTINUING DATA

This application is a continuation of U.S. patent application Ser. No. 17/214,759, filed Mar. 26, 2021. U.S. patent application Ser. No. 17/214,759 is a continuation of U.S. patent application Ser. No. 15/254,877, filed Sep. 1, 2016, (now issued as 10,987,129). U.S. patent application Ser. No. 15/254,877 claims priority from U.S. Provisional Patent Application No. 62/214,297, entitled "Multi-Shield Spinal Access System" by Thommen et al., filed Sep. 4, 2015. The entire contents of each of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Today, microsurgical spinal bone resections and spinal decompressions which are performed under microscopic view through mini-open tubes and retractors are becoming the standard of spinal surgical care. These access tools normally have inner diameters between about 16 mm and 30 mm. Where, as here, the approach and decompression technique are familiar to spinal surgeons, and where standard equipment and instruments can be used, these known technologies should be considered as a base from which further innovation can be made.

However, the anatomic window of Kambin's triangle, through which safe disc access is possible, has very limited dimensions. This access window can be enlarged by resecting at least a part of the superior articular process. But either way, the length of a working shield needed to safely introduce the implant to the intervertebral space via this approach must be in the region of about 8-12 mm in diameter, reaching from the facet joint line to the disc entry point.

SUMMARY OF THE INVENTION

The present inventors envision introducing a second, inner shield through the above-mentioned first, outer shield. The second inner shield extends past the first outer shield to arrive next to nervous tissue, thereby shielding the nerves from instruments or devices passing through to the disc space. During this step, the outer shield allows the visual, safe placement of the inner shield.

In one embodiment, there is provided an outer shield (which can be, for example, a tube or a blade) comprising an access shield with a larger diameter (~12-30 mm) that reaches from the skin down to the bone line, with an inner shield having a second smaller diameter (~5-12 mm) extending past the access shield and reaches down to the disc level. This combines the benefits of the direct visual from microsurgical/mini open approaches and percutaneous techniques (FIGS. 1a-1b and 2).

The outer shield has a number of features and advantages. First, it enables separation and protection of surrounding of soft tissue and visualization during a standard microsurgical decompression/bone resection work under microscopic view based on a standard procedure that a surgeon who is familiar with MIS techniques is able to perform. Second, it enables separation and protection of surrounding of soft tissue and visualization during detection and removal of the facet joint, or parts of the facet joint based on a standard procedure that a surgeon who is familiar with MIS technique is able to perform. Third, it enables identification, preparation and protection of sensitive (e.g., neural) tissue (exiting nerve root, traversing nerve root, dura) under direct visual control

underneath the border between retraction-sensitive and non-retraction sensitive tissues (e.g., the facet line) based on a standard procedure that a surgeon who is familiar with MIS technique is able to perform. Fourth, it enables insertion of the inner shield and potential docking of the inner shield in the disc space or at the vertebrae under direct visual control.

Likewise, the inner shield has a number of features and advantages. First, it enables protection of nervous tissue (exiting nerve root, transverse nerve root, dura) against instruments that are introduced into the intervertebral disc. Second, it enables guidance of intradiscal instrumentation (discectomy instruments, visualization instruments, discectomy verification instruments, implant insertion instruments, bone graft filling instruments). Third, because of its small size, the shield can be inserted with minimal damage or trauma to bone and soft tissue in the area of the posterior column of the spine, comparable to percutaneous access instruments.

Therefore, in accordance with the present invention, there is provided a method of accessing an intervertebral disc in a patient, comprising the steps of:

- a) making an incision in a skin of the patient,
- b) percutaneously inserting through the incision an outer shield having a substantially tubular shape (such as a tube or a multi-slotted retractor), the outer shield having a length adapted to extend from the incision to a border between sensitive and insensitive tissue (e.g., a superior articular process (SAP), or a lamina), in the spine of the patient,
- c) stabilization of this outer shield to a pedicle anchor,
- d) insertion of an outer shield integrated optical visualization instrument,
- e) resecting a portion of the superior articular process, and/or performing a microsurgical decompression procedure
- f) inserting or deploying an inner shield through or from the outer shield so that a distal end portion of the inner shield extends to the disc, the inner shield having an outer surface,
- g) contacting the outer surface of the shield to a nerve root to shield the nerve root,
- h) microsurgical decompression of any tissue deemed to be causing nerve impingement,
- i) extraction of the intervertebral disc material including the removal of the cartilaginous material from the vertebral endplates,
- j) insertion of the interbody device, and
- k) deployment of a mechanism of stabilization to stabilize the intervertebral segment.

Also in accordance with the present invention, there is provided a method of accessing an intervertebral disc in a patient, comprising the steps of:

- a) making an incision in a skin of the patient,
- b) percutaneously inserting through the incision an outer shield having a substantially tubular shape,
- c) stabilization of this outer shield to a pedicle anchor,
- d) inserting an inner shield through the outer shield so that a distal end portion of the inner shield extends to the disc, the inner shield having an outer surface,
- e) contacting the outer surface of the shield to a nerve root to shield the nerve root
- f) microsurgical decompression of any tissue deemed to be causing nerve impingement,
- g) extraction of the intervertebral disc material including the removal of the cartilaginous material from the vertebral endplates,
- h) insertion of the interbody device, and

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i) deployment of a mechanism of stabilization to stabilize the intervertebral segment.

Also in accordance with the present invention, there is provided an access device for accessing an intervertebral disc, comprising:

- a) an outer shield having a substantially tubular portion, a length adapted to extend from an incision to a border between sensitive and insensitive tissue (e.g., an articular process or a lamina), a proximal end portion, a distal end portion, an outer surface, and a longitudinal through-bore defining an inner surface,
- b) an inner shield having i) a first substantially tubular portion having a proximal end portion, a distal end portion, a longitudinal through-bore defining an inner surface, and an outer surface defining a diameter, and ii) a longitudinal flange extending distally from the distal end portion of the substantially tubular portion, wherein the outer surface of the inner shield substantially nests within the inner surface of the outer shield so that the flange extends distally past the distal end portion of the outer shield.

DESCRIPTION OF THE FIGURES

FIG. 1*a* shows an interbody device delivered through the access device;

FIG. 1*b* shows an end view of the access device;

FIGS. 2*a* and 2*b* are different views of a tube-in-tube embodiment of the access device;

FIGS. 3*a-3d* show different axial cross sections of the inner shield;

FIG. 4 shows a necked, funnel-shaped embodiment of the inner shield;

FIGS. 5-6 show different longitudinal cross sections of concentric and nonconcentric inner shields;

FIG. 7 shows a jointed access device;

FIG. 8*a* shows a flanged embodiment of the inner shield;

FIG. 8*b* shows an inner shield with a proximal stop;

FIG. 9 shows an access device with two ports attached to the outer shield; one port is a connector to hold the outer tube, while the other is an interface for a light source;

FIG. 10 discloses a cross-section of an outer tube wherein the outer tube wall has a first channel adapted for containing a visualization unit (such as a camera) and a second channel adapted for containing a cleaning system (such as a lens cleaning device);

FIG. 11 discloses a cross-section of an outer tube wherein the outer tube wall contains a lens cleaning device and a camera;

FIG. 12 discloses a chip-on-tip embodiment including a cross-section of an outer tube wherein the outer tube wall has a channel containing an endoscope having a video chip near its distal end;

FIG. 13 discloses a distal end of an outer tube featuring a video chip near its distal end;

FIG. 14 discloses a scope holder for an endoscope;

FIGS. 15 *a-15b* show inner shield having proximal elbows;

FIG. 16 shows an access device with a distal sharpened tip on the inner shield within an outer shield;

FIG. 17 shows an access device with a positioning ring between the inner and outer shields;

FIG. 18 shows an access device with a depth adjustment means formed by the inner and outer shields;

FIG. 19 discloses an integrated retractor having a flat inner face housed within a cutout of an outer tube 245;

FIG. 20 shows the embodiment of FIG. 48;

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FIG. 21 discloses a retractor having a flat inner face housed within an outer tube;

FIGS. 22-24*b* show a distraction embodiment;

FIGS. 25-30 show an access device with an extending shield;

FIG. 31 shows an access device with an inner and outer shield;

FIGS. 32-34 show in inner shield;

FIG. 35 discloses a radial soft tissue retractor;

FIG. 36 discloses an outer tube/inner retractor assembly wherein the first inner retractor and second inner retractor both tilt inwards to retract soft tissue;

FIGS. 37-46 disclose a preferred method of surgery involving the access device;

FIGS. 47*a-47c* disclose a Navigation plug comprising a base having an array attached thereto, wherein the plug is adapted to fit within an outer tube;

FIG. 48 discloses the cookie cutter-type distal end of an ultrasonic cutter extending from the end of an outer tube, wherein the distal end has a plurality of cutting teeth;

FIGS. 49*a-49b* disclose various cross-sections of the template for guiding a bone cutting device;

FIG. 50 discloses a cookie cutter-type distal end of an ultrasonic cutter having a semicircular cutting piece cutter bone;

FIG. 51 discloses a mini flex arm connecting an outer tube and a screw extension;

FIG. 52 discloses an outer tube/inner retractor assembly wherein the inner retractor is tilted inwards to retract soft tissue;

FIG. 53 discloses an outer tube/inner retractor assembly wherein the inner retractor runs parallel with the outer tube; and

FIG. 54 discloses an endoscope housed within an outer tube, and an inner tube extending from the outer tube.

DETAILED DESCRIPTION OF THE INVENTION

Fluoroscopic visualization is performed to define the incision site of the initial reference array placement, as well as the incision for access to the intervertebral disc.

Generally, the shields of the present invention can be applied to any of the conventional approaches commonly used in spine surgery. However, given the clinical benefit of the access device and its underlying rationale, it is preferably suitable to use these shields in either interlaminar, extraforaminal or transforaminal approaches to the intervertebral disc.

Now referring to FIGS. 1-7, there is provided an access device for accessing an intervertebral disc, comprising:

- a) an outer shield 1 having a substantially tubular portion, a length adapted to extend from an incision to a border between sensitive and insensitive tissue (e.g., an articular process), a proximal end portion 3, a distal end portion 5, an outer surface 7, and a longitudinal through-bore 9 defining an inner surface 10,

- b) an inner shield 11 having i) a first substantially tubular portion 13 having a proximal end portion 15, a distal end portion 17, a longitudinal through-bore 19 defining an inner surface 21, and an outer surface 23 defining a diameter, and ii) a longitudinal flange 25 extending distally from the distal end portion of the substantially tubular portion,

wherein the outer surface of the inner shield substantially nests within the inner surface of the outer shield so that the flange extends distally past the distal end portion of

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the outer shield, and the distal end portion of the substantially tubular portion of the inner shield extends distally past the distal end of the outer shield.

Outer Shield Embodiments

In the design of the outer shield, traditional tube or split tube/retractor concepts can be used. Newer concepts such as a “flexible tube” could also be adopted. The outer shield can be a simple cylindrical tube. It may also be a split tube, in the manner that conventional retractors are considered to be split tubes. It can be a flexible tube. It can be a tube with a slot running from the proximal end to the distal end. Various shape embodiments could be:

- a) a cylindrical tube with an inner diameter D (FIG. 3a);
- b) an oval tube with a height a different than a length b (FIG. 3b);
- c) a “half moon” tube having a substantially circular or oval cross section of diameter D, with a section cut (or chord) “a” (FIG. 3c); and
- d) a rectangular tube with height a and width b (FIG. 3d).

In some embodiments, the shape of the distal end portion 5 includes an unsymmetrical shape for better tissue retraction lateral to the SAP.

The outer shield can be preferably used with a variety of access window sizes (i.e., widths) ranging from 6 mm to 25 mm and lengths ranging from 40 mm to 200 mm. Typically, the outer shield comprises a feature that allows for the attachment of a stabilization mechanism that allows for appropriate flexibility in attachment (e.g. a ball joint). In one embodiment, the outer shield has a customized feature adapted for the introduction of an endoscope or camera that allows the endoscope to be introduced to a predetermined depth where the working window at the distal portion of the outer shield can be visualized.

Inner Shield Embodiments

Now referring to FIGS. 3a-3d, the inner shield 11 may encompass various designs as well.

In a first embodiment, the inner shield is a fully surrounding (i.e., extending for 360 degrees) stiff tube. It may possess various cross-sections, such as:

- e) a cylindrical tube with an inner diameter D (FIG. 3a);
- f) an oval tube with a height a different than a length b (FIG. 3b);
- g) a “half moon” tube having a circular cross section of diameter D, with a section cut (or chord) “a” (FIG. 3c); and
- h) a rectangular tube with height a and width b (FIG. 3d).

The inner shield may possess different longitudinal shapes. For example, in a second embodiment, and now referring to FIGS. 4-7, the inner shield 11 is a funnel-shaped (e.g. necked) tube (as in FIG. 4). In this embodiment, it changes its cross sectional shape/area along the shield, with a bigger diameter/working zone at the proximal portion, and the length of this zone with a bigger diameter is adjusted to be the part of the inner shield that will be nested within the outer shield, and a smaller diameter/working zone where the inner shield is extending the outer shield. This design increases the range of motion of intradiscal tools and enables better visualization. In FIG. 4, the flange is a second substantially tubular portion 25 having a diameter less than the diameter of the first substantially tubular portion 13 of the inner shield. A necked region 27 is disposed between the first and second substantially tubular portions.

In some embodiments, the inner shield may be in the form of one of a plurality of retractor blades.

In tubular embodiments, the smaller tube can be a concentric with the larger tube, or not concentric therewith. In FIG. 5, the first and second substantially tubular portions of

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the inner shield are concentric ($a=b$). In FIG. 6, the first and second substantially tubular portions of the inner shield are not concentric ($a>b$).

In some embodiments, there is provided a spherical joint between the larger and the smaller tubes, allowing the angle to change between the two tubes (FIG. 7). In FIG. 7, the outer surface 23 of the inner shield substantially nests within the inner surface 10 of the outer shield so that the proximal end of the substantially tubular portion of the inner shield terminates within the outer shield. Also, the distal end portion of the inner shield narrows distally to define a first radius R1, and the proximal end portion of the inner shield narrows distally to define a second radius, and the proximal end portion of the inner shield nests within the distal end portion of the outer shield to allow polyaxial pivoting of the inner shield.

In some embodiments, the inner shield is a partially surrounding tube/shield, or “flange,” designed only to protect the nerves. For some applications, the only purpose of the inner tube might be to shield/protect the exiting nerve root. In this case, the inner shield might be simplified to a cylinder with a flange 25 extending distally therefrom, so that the flange is only a shield of about a quarter of a full circle. See FIG. 8a, or FIG. 9 if mounted on the outer shield. Depth Adjustment of Nerve Protector

The aforementioned outer shield can be positioned and fixed in its depth through a mechanism which relies on interference between the outer shield and the inner shield at any location along either the outer shield or inner shield.

In FIG. 8b, the proximal end portion of the first substantially tubular portion 13 of the inner shield comprises a stop 31 adapted to abut the proximal end portion of the outer shield, the stop being adapted to prevent excessive distal movement of the inner shield. Preferably, the stop extends substantially radially about the proximal end portion of the substantially tubular portion of the inner shield. The stop may also further comprise a textured radial surface 33 adapted for gripping. It acts as both a stop and as a handle to twist the shield.

In FIG. 9, the outer surface of the inner shield substantially nests within the inner surface of the outer shield so that the proximal end portion of the substantially tubular portion of the inner shield extends proximally past the proximal end of the outer shield. Also in FIG. 9, the outer surface of the outer shield further comprises a first port 41 adapted for connecting to a navigation instrument or a stabilization point, and a second port 43 adapted for connecting to a camera/light system.

Navigation of Outer Shield

The first port allows the outer shield to be navigated to determine its position (depth and orientation) in relation to the treatment site. In one embodiment, the outer surface of the outer shield has a feature that allows for the direct or indirect attachment of a navigation instrument. In another embodiment, the inner surface of the outer shield has a feature that allows for the direct or indirect attachment of a navigation instrument.

Endoscope in the Outer Shield

In some embodiments, the outer shield has an integrated endoscope that can be set in a fixed or variable (angle or circumferential) position relative to the anatomy. This endoscopic visualization can be utilized in subsequent surgical steps, including bone removal, inner shield deployment, discectomy, and implant insertion. Preferably, the endoscope has an integrated lens cleaning mechanism for automated lens cleaning in situ.

FIG. 10 discloses a cross-section of an outer tube wherein the outer tube wall **138** has a first channel **238** adapted for containing a camera and a second channel **338** adapted for containing a lens cleaning device.

FIG. 11 discloses a cross-section of an outer tube wherein the outer tube wall contains a lens cleaning device **139** and a camera **239**.

FIG. 12 discloses a chip-on-tip embodiment including a cross-section of an outer tube wherein the outer tube wall has a channel **140** containing an endoscope **240** having a video chip **340** near its distal end.

FIG. 13 discloses a distal end of an outer tube featuring a video chip **141** near its distal end.

Fixed Endoscope

The endoscope can be a chip-on-tip type of endoscope having an outer diameter less than 5 mm and having an incremental length substantially matching the length of the outer shield. The benefits of an integrated chip-on-tip endoscope/outer shield embodiment include the relatively free space within the bore of the outer shield, thereby enhancing visualization.

Preferably, the endoscope is angled within the port or has a built-in lens angle such that, at final positioning within the port, the circumference of the distal portion of the outer shield is visible and the area within the circumference is visible as well.

In some embodiments, the endoscope can be removed from the wall of the outer shield and inserted independently into the outer shield bore to inspect the treatment site (e.g. into the disc space for confirmation of adequate discectomy).

FIG. 14 discloses a scope holder **137** for an endoscope **237**.

Still referring to FIG. 9, in some embodiments, the flange **25** of the inner shield has an arcuate transverse cross-section. In some embodiments, the arcuate transverse cross-section of the flange defines an outer surface **47** of the flange having a curvature substantially similar to a curvature of the inner surface **10** of the outer shield. Preferably, the flange defines a second substantially tubular portion having a diameter less than or equal to the diameter of the first substantially tubular portion of the inner shield.

Now referring to FIGS. **15a-15b** and **16**, the inner shield can be a single blade that can be mounted/hooked to the outer tube. In this case, elbow **51** functions as a stop and also as a connector to the outer tube. Also, the proximal end of the inner shield may form an anchoring spike **53**.

There are a number of ways to fix or locate the inner shield onto the disc and/or onto the outer shield. In one embodiment, which provides safety of the inner shield against slippage/dislocation, involves mounting it distally (onto or within the vertebral endplates or disc annulus) and/or proximally (onto the outer shield).

Distal fixation of the inner shield with the anatomy may include: a) fixation within disc annulus; b) fixation against vertebrae; c) fixation against other structures; d) K-Wires that are distally extending through the walls of the inner shield and anchored to the anatomy; and e) spikes extending the distal part (FIG. 15) to be anchored to the anatomy.

Proximal fixation of inner shield upon the outer shield may involve a positioning ring or a depth adjustment. Now referring to FIG. 17, proximal fixation of inner shield upon the outer shield may involve a positioning ring **55**. Assuming the outer shield would be fixed relative to the anatomy, there would be the option of having positioning rings having the shape of the outer tube at the outside, and of the inner tube on the inside. When placed over the inner shield and into the

outer shield, such a ring would stabilize the location or at least the orientation of the proximal inner shield against the outer shield, and—by considering the assumption above—also against the anatomy.

Now referring to FIG. 18, proximal fixation of inner shield upon the outer shield may involve a depth adjustment means **57**. This would additionally stabilize or anchor the tip location of inner shield against the anatomy via anchoring or hooking the inner shield into the outer shield via ratchet system. The ratchet system can also be located between the inner surface of the outer shield, and the outer surface of the inner shield or within the wall of the outer shield. It may further include a spring system to increase friction between the inner and outer shields.

Inner Shield Deployment (Circumferential)

Embodiments having separate outer and inner shields allow for the independent positioning of the inner shield relative to the outer shield. Also, the use of a smaller inner shield (relative to the outer shield) allows for maximum visualization at the entrance where no retraction-sensitive tissues reside. This maximum visualization allows for accurate placement of the inner shield. Where retraction-sensitive tissues reside distal the outer shield, a relatively smaller inner shield allows for minimum retraction while providing an access through or past these tissues. Preferably, the inner diameter of the inner shield is no more than 40%-100% of the inner diameter of the outer shield.

In some embodiments, the inner shield-outer shield configuration is replaced by a) a primary shield having a substantially tubular shape having a cutout, and b) a secondary shield having a shape that is substantially insertable into the cutout. Preferably, the primary shield has a substantially annular shape and the secondary shield has an arcuate cross-section that substantially matches the annular shape of the primary shield. This embodiment allows the secondary shield to be tilted with respect to the primary shield.

Inner Shield Deployment (Radial)

In another nerve protection embodiment, the motion of retraction of the shields is radial rather than rotational. In these embodiments, a straight or bayonetted inner shield may be used. The inner shield may be positioned over the area in which the protected tissue is to be located. The flange shield can then be angled into the center of the access window at the distal end of the outer shield e.g. towards the caudal pedicle. It can then be subsequently advanced longitudinally onto the medial side of the nerve root, into the “safe zone” as described by Kambin. It is subsequently angled such that the distal tip of the inner shield is angled laterally, wherein its outer distal surface gently pushes the existing nerve root away and/or shields it against the tools that are further introduced medially to the shield for intradiscal work. This embodiment may be constructed such that the inner shield substantially nests either a) within the wall of the outer shield (FIG. 19(a)), b) inside the inner surface of the outer shield (FIG. 20(b)), or c) outside the outer surface of the outer shield ((FIG. 21(c)). In some embodiments, the inner shield is built into the wall of the outer shield or even outside the outer shield.

FIG. 19 discloses an integrated retractor having a flat inner face **145** housed within a cutout of an outer tube **245**.

FIG. 20 discloses a retractor having a flat inner face **144** housed within an outer tube.

FIG. 21 discloses an outer tube **150** having a retractor **250** nesting with the outer face **350** of the outer tube.

Depth Control of Nerve Protector

The aforementioned outer shield can be controlled in its depth through a mechanism that relies on interference

between the outer shield and the inner shield at any location along either the outer shield or inner shield.

There are a number of avenues by which the present device can be used to distract the disc space and/or provide nerve protection upon mounting.

In one distraction embodiment, a revolution spreader is used. This is a conventional concept involves an ovoid or rectangular cross-sectional shaped rod that is inserted into the disc with its smaller dimension directed towards the vertebral endplates. After turning the spreader by 90° under force, the larger dimension is directed towards the vertebral endplates, which distracts the disc by the difference of the two cross sectional dimensions.

In a second distraction embodiment, as now referring to FIGS. 22-24b, the inner shield may comprise a spreader, which includes a frame 60, a cranial blade 61, and a caudal blade 63.

The spreader with respective cranial 61 and caudal 63 distraction blades in cranial and caudal locations is introduced into the disc in a collapsed/tapered configuration (FIG. 22). The spreader blades are then distracted with an inner core 65 (the core matching a counter geometry on the blade to not slip away sideways), elevating the intervertebral height from d1 to d2 (FIGS. 22-23). The side walls 67 matching to the inner core height are then introduced medially/laterally (FIG. 24a), to circumferentially close the four-wall shield. Once the inner core is removed, the stacked shield keeps the vertebral bodies separated in distracted condition (FIG. 24b).

Now referring to FIGS. 25-26, the inner shield may further comprise a rotating flange 71 that moves laterally/medially upon rotation to shield the nerve root.

In a nerve protection embodiment, and now referring to FIGS. 27-30, a rotation funnel 70 is used. Preferably, the flange shield 71 can be smartly introduced to protect the exiting nerve root while being inserted. This shield can be directed towards the caudal pedicle if introduced through the outer shield. This location is a "safe zone." Once the distal tip reaches the disc level, the inner shield can be turned clockwise by about 90° (i.e., rotated), so that the flange gently pushes the exiting nerve root away, and/or shields it against the tools that are further introduced medially to the shield, for intradiscal work.

In a second nerve protection embodiment, and now referring to FIGS. 31-34, a concentrically-arrayed multi-shield is used to gently move and/or shield nerves. The rotation funnel principle can also be applied for more than one rotating shield. A single shield may be suitable if the protection only has to be provided against a structure that lies on one single side. In other situations, however, the shield entry towards the disc would be bounded both medially and laterally by the traversing and the exiting nerves, so the inner shield needs to shield against two opposing structures. In this case, the two concentrically-arranged outer 81 and inner 83 rotating flanges are turned by 90° in respective clockwise and counterclockwise directions to reach an end configuration wherein the opposed shields protect the nerves from the tools that are further introduced for intradiscal work.

In another nerve protection embodiment, a radially-retracting multi-shield is used to gently move and/or shield nerves. The radially-retracting principle can also be applied to more than one radially retracting shield.

FIG. 35 discloses a radial soft tissue retractor 136.

FIG. 36 discloses an outer tube/inner retractor assembly wherein the first inner retractor 153 and second inner retractor 253 both tilt inwards to retract soft tissue.

A single shield may be suitable if the protection only has to be provided against a structure that lies on one single side. In other situations, however, the shield entry into the disc would be bounded both medially and laterally by the traversing and the exiting nerves, so that the inner shield needs to shield against two opposing structures. In this case, the two opposing inner flanges are initially positioned towards the center of the outer tube access window and subsequently retracted outwards to shield the opposing nerves from the tools that are further introduced for intradiscal work.

FIGS. 37-46 disclose a preferred method of surgery involving the tube-in-tube access device.

In one embodiment, and now referring to FIG. 37, the surgeon places a pedicle screw-based anchor, adds a navigation reference frame 101 to the anchor, and uses a commercial navigation system for navigation. In some embodiments, a navigation array is placed onto the anatomy with reference to an anatomical feature that is symmetrically substantially adjacent the treatment site (e.g. contralateral cranial or caudal pedicle).

In some embodiments, there is navigation of the probe to a facet capsule or disc space through Kambin's triangle. Preferably, subsequent to fascia and muscle dissection, a probe enabled with navigation visualization is introduced to achieve an initial anchoring point. In one embodiment, the probe is inserted into the disc space by being indexed off the lateral border of the superior articulating process and may be optionally enabled with/supported by a nerve detection and/or visualization function. In another embodiment, the probe is introduced into the facet capsule.

In some embodiments, there is dilation over a navigated probe. Subsequent to the initial anchoring point, dilation is performed to prepare the surgical site for the size of port required to perform the treatment. Sequential dilation up to the preferred size port window is then performed. The port is then introduced over the associated dilator. In one embodiment, the initial anchoring is in the disc space and concentric sequential dilation device(s) would be used in order to retract tissue concentrically around the initial anchoring point (exposing the lateral portion of the SAP on the lateral aspect and Kambin's triangle on the medial aspect). In another embodiment, the initial anchoring is in the facet capsule and eccentric sequential dilation device(s) could be used to focus tissue retraction laterally over the lateral portion of the SAP and Kambin's triangle.

In some embodiments, the outer shield is stabilized onto an anatomical reference. The outer sleeve has a substantially tubular portion having a point or feature designed for attachment to a stabilization mechanism, which in turn is fixed to an anatomical feature on the vertebral body either cranial or caudal to the treatment site.

FIG. 38 discloses an outer tube into which a plug containing a template for guiding a bone cutting device.

In some embodiments, the outer sleeve is attached to a stabilization mechanism. In one embodiment, this stabilization device would be a device of sufficient length to reach an anatomical fixation point (e.g. pedicle screw) on the contralateral side of the treatment site. The mechanism (including its connection feature connecting to both the outer shield and the anatomical anchor) allows for sufficient flexibility of placement of the outer shield and sufficient stabilization to hold the outer shield in place until it is released by the user. The method of stabilization would be such that the user can dictate the degree of stiffness.

In another embodiment, this device has sufficient length to reach an anatomical fixation point (e.g. pedicle screw) on the ipsilateral side of the treatment site. Likewise, the mecha-

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nism (including its connection feature to both outer shield and anatomical anchor) would allow for sufficient flexibility of placement of the outer shield and sufficient stabilization to hold the outer shield in place until released by the user. The method of stabilization would be such that the user can dictate the degree of stiffness.

In another embodiment, this device would be a device of sufficient length to reach an anatomical fixation point (e.g. pedicle screw) on midline of the patient. Likewise, the mechanism (including its connection feature to both outer shield and anatomical anchor) would allow for sufficient flexibility of placement of the outer shield and sufficient stabilization to hold the outer shield in place unless released by the user. The method of stabilization would be such that the user can dictate the degree of stiffness.

Now referring to FIGS. 39a-39b, the surgeon then dilates the tissue superior to the pedicle-based anchor, and inserts an outer shield 1, connected to the anchor, with its proximal end directed to the superior articular process. Blunt dissection up to the bone is carried out over the affected intervertebral disc, and muscle retraction over the affected intervertebral disc is then carried out. This retraction involves blunt dissection of the muscle and fascia to bone level under direct visualization.

Now referring to FIG. 40, the surgeon then turns the outer shield 1 to the interlaminar space, preforms a central, bilateral decompression as required by the pathology, and then turns the shield back to its original position.

In some embodiments, an alternative to angling the access channel medially from the incision site could be the use of an alternative access site that would be more medial. In some embodiments, the initial anchoring point in the disc space will be medial to the inferior articulating process. For the embodiment having an initial anchoring point in the facet capsule, the dilation of the eccentric dilators will be medial from the capsule. Also, portions of the lamina and the inferior articulating process will be removed through the bone removal segment.

Now referring to FIG. 41, the surgeon then inserts a bone removal tool (not shown) into the outer shield tube and resects the lateral portion of the superior articular process to medially extend the traditional Kambin's triangle.

Under either direct or endoscopic visualization, a bone removal device is introduced to the outer shield and utilized to remove at least the lateral portion of the SAP. Such a device is available in lengths and sizes allowing for its safe introduction and use through an access window from 40 mm to 200 mm and a window size from 10-25 mm.

In one embodiment, this bone removal device is an ultrasonic cutting device. In another embodiment, this bone removal device is a reciprocating cutting surface. In yet another embodiment, this bone removal device is a revolving cutting tool. In another embodiment, this bone removal device is a mechanical punch with a stroke length between 10-30 mm. Removal of the bone can be performed in such a manner that sizes smaller than the access size will be excised and removed. The bone removal can be performed with the use of a template independently inserted into the outer shield and used to guide the direction of bone cutting and removal.

A Negative Template is a plug-like device that is inserted in the outer Access Tube. It contains a longitudinal cut-out in different shapes, depending on the cross-sectional shape of the tissue that needs to be removed respective of the cross-sectional shape of the tissue that needs to be covered and therefore protected from any surgical interactions. By inserting a cutting device like, e.g., a Milling Bit, into the

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longitudinal cut-out the surgeon is able to remove the tissue without the risk of endangering the covered tissue/structures. In combination with a proximal stop-system (on proximal end of outer Access Tube and/or shaft of milling system) the surgeon can remove the tissue layer by layer. The layer thickness and therefore the progression of the cutting procedure can be controlled via the stop system supported by a scale. This system allows the surgeon to perform safe tissue removal with a controlled serial work flow: check anatomical situation→adjust stop system to define cross-sectional thickness of tissue that needs to be removed→insert milling system until the stop system is engaged→mill/cut tissue (also blindly) in plane (2D)→remove milling system→check anatomical situation→adjust stop system.

A serial workflow can be considered to be safer than a parallel workflow, since the surgeon only needs to take care of one parameter at a time (here: planar position of milling bit followed by its depth followed by planar position of milling bit . . .) whereas a parallel workflow requires the control of two or more parameters at a time (here: planar position of milling bit in parallel to its depth).

Navigation of SAP Removal can be carried out with the aforementioned bone removal device adapted to be navigated through its mechanical or visual connection with a navigation system.

Now referring to FIGS. 42-43, the surgeon then inserts the inner shield tube into the outer shield tube, which acts to extend the outer tube anteriorly from the facet line until the tip of the inner shield reaches the level of the disc. The nerve root is protected by the inner shield.

Now referring to FIG. 44, the surgeon then identifies the disc, spreads the disc with a wedged osteotome; checks the mobilization, and removes the posterior rim, osteophytes and annulus until a minimum annular window is opened. The surgeon then inserts a disc removal tool 201 into the access device, removes the disc and prepares the endplates.

An alternative embodiment to the prescribed disc clearing step in FIG. 44 would be to have the disc removal tool navigated through its mechanical or visual connection with a navigation system.

Now referring to FIG. 45, the surgeon then performs temporary disc space distraction, fills parts of the disc space with bone graft and inserts a fusion cage 203 into the remaining disc space.

Now referring to FIG. 46, the surgeon then adds posterior fixation 103.

Viewing Element

In some embodiments, a visualization element based on the chip-on-tip technology and integrated into the wall of the port is used. This embodiment has a number of advantages over a standard rod-lens endoscope that is mounted at the tube wall:

Manufacturing costs. The chip-on-tip technology allows a very cost efficient manufacturing, therefore can be marketed as a 'single use' instrument.

Rigid portion only at distal tip. Whereas a standard rod-lens endoscope system has a stiff, cylindrical shape throughout the whole tube, the chip-on-top endoscope may have a non-cylindric configuration at the proximal outer tube end. Preferably, this shape is a flat cable shape. In some novel embodiments, in relation to a standard rod-lens-endoscope, the chip-on-tip endoscope has a relatively short "stiff" section (about 20 mm), where the proximal portion consists of a cable that can be flexible. In other embodiments, the stiff portion is shorter (producing a smaller chip-assembly) and actively articulating concepts are used to

change the lens angle. Due to the cable's integration in the tube wall, the shape of the port window is maintained throughout the procedure. For example, a 5 mm chip on tip endoscope turns a 15 mm circular access window into a kidney shaped access window.

Size/weight of camera unit. A standard rod-lens endoscope has a standard eyepiece that is a universal interface with a certain size. The camera that is connected to such a system has to be built in a certain dimension to be compatible with the eyepiece. This requirement produces a relatively bulky camera attachment (approx. 3-6 cm in diameter, approx. 5-10 cm in length) having a number of drawbacks. First, this large camera construct can be a physical obstacle to work, especially if the trajectory of the working port changes or interferes with the camera. Secondly, the dimension and weight of this conventional construct becomes significant enough to produce certain undesirable forces upon the rod-lens-endoscope, especially in bending. Thirdly, the relatively fragile conventional rod-lens-endoscope has to be embedded in stabilizing structures such as metal tubes, thereby further reducing the active working window.

With the chip-on-tip embodiments disclosed herein having its chip cable embedded in the wall of the outer tube, the cable that exits at the proximal outer tube wall does not produce similar forces upon the working port. Also, respecting the attachment mechanisms that mount the chip-on-tip endoscope in the tube-wall, the lack of bending forces produced thereby raise the possibility of adopting relatively thin attachment options that mechanically do not need to be very stable.

Working Environment. Conventionally, a constant fluid environment (permanent flow of saline solution) is used in spine endoscopy applications. However, in a mini-open and microsurgical environment, the fluid environment is not helpful, as the anatomical conditions are very different. Accordingly, in the preferred novel procedures described herein, the chip-on-tip endoscope works in a dry, open air environment. However, the open, dry air environment in which the chip-on-tip endoscope is used may produce an undesired condensation effect upon the lens component of the endoscope. For example, a colder lens in a humid body temperature environment may fog up. Moreover, drill debris, burr debris or smoke from monopolar scalpels or hemostatic tools can likewise affect the lens of the endoscope so as to reduce visibility. Accordingly, it may be desirable to periodically clean the lens of the chip-on-tip endoscope.

Nerve Deflection (Tube in Tube)

In minimally invasive spine surgery conducted through portals, a set of dilators is often used to prepare the site for reception of the portal. One such technology is shown in US Patent Publication US 2012-0232552 (Morgenstern). In this conventional technology (which has eccentric dilators), the outer diameter of any one of the dilators is identical to the inner diameter of the next successive (outer) dilator. This identity of diameters is necessary for fluoroscopy assisted, percutaneous muscle dilation.

Since some embodiments of the present application describe a procedure between the level of the facet joint and the disc, the surgical site is dissected under direct visualization. Accordingly, the diameters of successive dilators used in these novel procedures do not have to match. Relaxation of the "exact diameter" requirement in these novel procedures allows the surgeon freedom in many tube design areas. For example, it allows the use of tubes that are tapered. It also allows the surgeon the freedom to use outer and inner ports that are not coaxial. It further allows the

trajectories of the inner port relative to the outer port to vary in angulation within certain treatment steps. Lastly, it allows the trajectories of the inner port relative to the outer port to vary in distance within certain treatment steps.

Because fluoroscopy-assisted, percutaneous muscle dilation is carried out without direct visualization, it is a blind procedure whose use has limitations. These limitations include the inability to carry out surgical steps that require direct visualization out of safety considerations. One such treatment step requiring direct visualization is direct decompression of bony and ligamentous tissue that is directly adjacent to nerve structures. Since this decompression is a common step performed on degenerated spinal motion segments during a spinal fusion, the Morgenstern procedure severely limits the pathologies/indications that can be treated.

Because some embodiments described herein allow for direct visualization of delicate anatomical structures, those embodiments further specifically allow direct decompression of bony and ligamentous tissue that is directly adjacent to nerve structures and more generally allow manipulation or removal of tissue adjacent the tubes through a very tissue-preserving "tube-in-tube" access port.

Morgenstern further describes a method in which a guide wire is directly introduced through the disc space to Kambin's triangle, under fluoroscopy guidance (i.e., no direct visualization). This is a well-known approach, but still requires inserting a guidewire past nerve structures without direct visualization. Morgenstern further describes the possibility of using electrically-based nerve monitoring probes. Moreover Morgenstern describes a method of enlarging the spinous process by subsequently rasping away bone from the SAP and the pedicle. Such a procedure might weaken the base of the pedicle, which makes it risky to combine this method with pedicle screw constructs.

The novel procedures described herein only perform non-visualized procedures (e.g., dilation) in a safe zone above the facet line. In the anatomically more critical zone between the level of the facet joint and the disc, the novel procedures dissect the surgical site under direct visualization, thereby allowing the surgeon to spare as much of the bone as is possible and as is meaningful.

Navigation

Navigation enhances static x-ray, CT, or MRI data by intra-operatively showing in real-time where the instruments used actually are in relation to the anatomy of the patient. Therefore it increases the safety of those instruments by showing their shape, trajectory and positioning and even more importantly it supports the surgeon to keep instrument orientation during the performed manipulations.

Without wishing to be tied to a theory, it is believed that one reason why minimally invasive techniques are not often used is the significantly higher x-ray exposure needed to keep orientation in comparison to mini-open techniques, where the surgeon still has direct visualization and so can actually see the active site with a microscope or loupe. The x-ray exposure is an even greater for the surgeon who is exposed to the radiation on a frequent basis.

This challenge is addressed by the implementation of navigation technology in the novel procedures described herein because they allow the reduction of x-ray exposure to an ideal minimal total of two x-rays for registration purposes. Once a single lateral shot and a single anterior-posterior shot have been registered, all used instruments (e.g. Jamshidi-Needle, Pointer, Dilators, Access Tube, Osteotome, Expandable Cage itself, Disc Removal Device . . .) can be projected in these static fluoro-images in real

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time. Another positive effect is a significant savings of time. Having the navigation system in place also helps the surgeon to understand the orientation (trajectory and depth) of the endoscope and therefore to understand what he or she actually sees with the camera. This can either be achieved by navigating the camera directly or indirectly by setting the camera in a fixed position integrated into a navigated Access Tube.

The Jamshidi-Needle, Pointer, Dilators, and Access Tube Instruments can all be navigated with only one Instrument, the FOX-Navigation-Multi-Tool.

FIGS. 47a-47c disclose a Navigation plug comprising a base 147 having an array 247 attached thereto, wherein the plug is adapted to fit within an outer tube 347. Bone Cutter

In some embodiments, the novel procedures use an Ultrasonic Bone Cutting device for SAP removal, which specifically cuts bone only and will not cut soft tissue. Embodiments based on a conventional Expandable Cage Device for interbody fusion may require an access window at least as large as 12 mm. Such a large window can only be achieved by (partly) removing the Superior Articulation Process (SAP) to extend the Kambin's Triangle. The Ultrasonic Bone Cutting Device adds significantly to the safety of this procedure since it does not cut nerves if accidentally hit. If the cutting device blade is designed to be in the shape and diameter of the Inner Tube/Blade (i.e., a Cookie Cutter design) that approaches distally down to the level of the disc space, the SAP removal can be minimized (less trauma, less stress for patient, quicker recovery) and performed in a single step (faster than multiple step procedure).

FIG. 48 discloses the cookie cutter-type distal end 143 of an ultrasonic cutter extending from the end of an outer tube 243, wherein the distal end 143 has a plurality of cutting teeth 343.

Another option to increase the safety of bone cutting is a depth-controlled manual milling of the bone with a negative guide. The negative guide covers those areas that will not be removed (negative template). The depth control allows the milling of the bone layer by layer, under serial control of the surgeon. The reference for the depth control as well as the trajectory can be the outer Access Tube (also see paragraph navigation).

FIGS. 49a-49b disclose various cross-sections of the template for guiding a bone cutting device. Bone Cutter

In some embodiments, the bone removal device is a harmonic scalpel having a cookie cutter design. The scalpel has a crescent-shaped cutting surface that interfaces with the outer tube. The scalpel is used as a single pass instrument, removing a predetermined amount of bone in a single pass. In some embodiments, the scalpel also has a tube that sprays water for irrigation, while the outer tube has a suction line for clearing the slurry of removed bone.

In some embodiments, the scalpel can be navigated and ride down a slot provided in the inside wall of the outer tube. The slot depth can be predetermined to provide depth-controlled milling, and to control where cutter goes. This is advantageous because it is believed that freehand cutting hits the nerves too easily. The shape and size of the cutting surface can define the specific area of bone to be removed. The specificity of cut is advantageous because it minimizes the amount of bone removal, which is beneficial in the highly enervated facet. Thus, a quicker procedure, less trauma (less pain), and more stable construct is realized. Viewing Element (Olive)

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In some embodiments, the chip viewing element can be angulated so it can see around a corner of the tube.

In a conventional endoscope, visualization is 2D (i.e., no depth perception), and so two nerves may look close together when they are actually 2 cm apart. Thus, in some embodiments, the endoscope is modified so that the chip acts like a range finder. In particular, the chip identifies and assesses a reference feature that is a known distance from chip, and then measures how far away a nerve is from chip (which is the tube end) based on that assessment. Nerve Deflection (Tube in Tube)

In some embodiments, the outer shield has a pressure sensor thereon to measure the stress on the nerve. Using ultrasound techniques that can measure distance, the system can measure the elongation of nerve under retraction and define a maximum elongation limit (e.g., 20%), and then warn the surgeon if the elongation limit is exceeded. In some embodiments, the system integrates ultrasound into the port and thereby navigates the port.

In some embodiments, the surgeon navigates the camera. This allows the surgeon to understand orientation of the camera.

In some embodiments, visualization provides an axial view of the disc, so the surgeon can understand the location of the disc removal tool.

Neuromonitoring Analytics

Currently, neuromonitoring devices can be used to obtain an indication of potential nerve health or nerve damage, which may be induced in a surgical setting. This indication of nerve health is achieved by measuring electrical impulses between a nerve near a surgical site and a far end of the nerve. For example, impulses may be measured between a nerve root at the spine and some point found on the legs.

Nerve damage can be caused through direct manual contact with a nerve. Apart from gross damage such as severing or crushing the nerve, other lesser forces imparted on the nerve can also cause damage. For example, displacing the nerve, stretching it, or compressing it can cause significant damage. In some cases, extended application of such forces to the nerve can reduce blood flow through the nerve, again causing nerve damage. Often times, this exposure time is dependent on the amount of force applied. Accordingly, there appears to be no known steadfast rule as to how long the surgeon may be able to load a nerve.

Alternate forms of evaluating potential nerve damage besides neuromonitoring may bring new insights into nerve protection during a procedure. In this regard, nerve manipulation measurement could yield an indication of risk to the nerve. If a nerve is displaced for the procedure, it may be elongated or it may be displaced laterally. These alterations in the nerve's physical features could be measured and used to predict potential nerve damage. Accordingly, other potential features could be measured and used to predict potential nerve damage include arc length and the diameter of the nerve itself etc. These features may be measured in quantifiable terms via techniques such as ultrasound. The resulting measurements are and then analyzed (via software or manually) in terms of absolute value, percent change or some other metric indicative of potential nerve risk/damage that can be obtained from a database or library. In some embodiments, these metrics can be used as predictors of the safe length of time that a nerve can have a given displacement or deformation without causing long term damage. Calculation or algorithms can also be used to determine a maximum safe deformation, or a maximum allowable time during which a nerve can have a given deformed feature.

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This measurement could be obtained in many ways. It can be measured manually, optically or through some other form of imaging. This could occur in an open procedure, subcutaneously in an MIS or other type of procedure. Direct visualization could be completed with the use of a camera. Before and after images could be interpreted to calculate the amount of absolute deformation or percent change. The measurements can be obtained through modalities such as ultrasound, or other forms of imaging that can “see” soft tissue or identify nerve tissue relative to the surrounding tissue (X-ray, CAT/PET scan, MRI, etc.).

In some neuromonitoring embodiments, it may be possible to measure impedance within the nerve or impulses, wherein this may be done locally relative to the specific deformation area of the nerve. In particular, in some embodiments discussed herein, the nerve shield could have a sensor on opposite edges of the shield that would contact the same nerve in two different nerve locations. These sensors would allow the surgeon to read electrical values such as impulses or resistance, before nerve distention and then measure it again as distention occurs or is achieved. The difference in these measured values could be an indicator as to the level of deformation.

FIG. 50 discloses a cookie cutter-type distal end 142 of an ultrasonic cutter having a semicircular cutting piece cutter bone.

FIG. 51 discloses a mini flex arm 146 connecting an outer tube 246 and a screw extension 346.

FIG. 52 discloses an outer tube/inner retractor assembly wherein the inner retractor 151 is tilted inwards to retractor soft tissue.

FIG. 53 discloses an outer tube/inner retractor assembly wherein the inner retractor 152 runs parallel with the outer tube 252.

FIG. 54 discloses an endoscope 154 housed within an outer tube, and an inner tube 254 extending from the outer tube.

In many embodiments disclosed above, an inner shield nests within an outer shield. In an alternative embodiment to all such embodiments, however, the inner shield is replaced with a removable blade that is integrated into a cutout formed within the wall of the outer shield. In such cases, the outer surface of the inner shield substantially nests within the outer surface of the outer shield so that the flange extends distally past the distal end portion of the outer shield.

In many embodiments disclosed above, the proximal end portion of the substantially tubular portion of the inner shield comprises a stop adapted to abut the proximal end portion of the outer shield, the stop being adapted to prevent excessive distal movement of the inner shield. In other embodiments, the abutment occurs anywhere along the outer shield.

We claim:

1. An access device for accessing an intervertebral disc, comprising:

- a) an outer shield having a length adapted to extend from an incision to an articular process, a proximal end portion, a distal end portion, an outer surface, and a longitudinal throughbore defining an inner surface,
 - b) an inner shield having a proximal end portion and a distal end portion; and
 - c) a video chip disposed in the distal end portion of the outer shield;
- wherein the inner shield is configured to be disposed within longitudinal throughbore of the outer shield

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such that the distal end portion of the inner shield extends distally past the distal end portion of the outer shield; and

wherein the video chip is disposed in a channel of the outer shield and the channel opens into the inner surface of the outer shield at least at the distal end portion of the outer shield.

2. The device of claim 1, wherein the channel intersects with the longitudinal throughbore of the outer shield along the distal end portion of the outer shield proximal to a distal end of the outer shield.

3. The device of claim 1, further comprising a flat cable coupled to the video chip and extending from a proximal end portion of the outer shield.

4. The device of claim 1, wherein the proximal end portion of the inner shield comprises a stop adapted to abut the proximal end portion of the outer shield, the stop being adapted to prevent excessive distal movement of the inner shield.

5. The device of claim 4 wherein the stop extends substantially radially from the proximal end portion of the inner shield.

6. The device of claim 1, wherein the distal end portion of the inner shield has an arcuate transverse cross-section.

7. The device of claim 6, wherein the arcuate transverse cross-section of the distal end portion of the inner shield defines an outer surface having a curvature substantially similar to a curvature of the inner surface of the outer shield.

8. The device of claim 6, wherein the inner shield has a proximal elbow that functions as a stop.

9. The device of claim 6, comprising a camera disposed in the channel of the outer shield, the camera having the video chip.

10. A method of accessing an intervertebral disc in a patient, comprising:

- a) making an incision in a skin of the patient,
- b) percutaneously inserting through the incision an outer shield having longitudinal throughbore, the outer shield having a length adapted to extend from the incision to the spine of the patient,
- c) inserting an inner shield within the longitudinal throughbore of the outer shield so that a distal end portion of the inner shield extends distally past the distal end portion of the outer shield and to the disc,
- d) disposing a video chip in a distal end portion of a channel of the outer shield, wherein the channel opens into the inner surface of the outer shield at least at the distal end portion of the outer shield.

11. The method of claim 10, further comprising resecting at least a lateral portion of the superior articular process.

12. The method of claim 10, wherein the channel intersects with the longitudinal throughbore of the outer shield along the distal end portion of the outer shield proximal to a distal end of the outer shield.

13. The method of claim 10, wherein the distal end portion of the inner shield has an arcuate transverse cross-section.

14. The method of claim 10, further comprising contacting the outer surface of the shield to a nerve root to shield the nerve root.

15. The method of claim 10, further comprising coupling a light source to the outer shield.

16. The method of claim 10, wherein the channel opens into the inner surface of the outer shield along the distal end portion of the outer shield proximal to a distal end of the outer shield.

17. The method of claim 10, further comprising contacting the outer surface of the shield to a nerve root to shield the nerve root.

18. The method of claim 10, wherein the inner shield is angled into a center of an access window at a distal end of the outer shield, and then is subsequently advanced longitudinally onto a medial side of a nerve root.

19. A method of accessing an intervertebral disc in a patient, comprising the steps of:

- a) making an incision in a skin of the patient, 10
- b) percutaneously inserting through the incision an outer shield having a substantially tubular shape, the outer shield having a length adapted to extend from the incision to the spine of the patient,
- c) stabilizing the outer shield to a pedicle anchor 15
- d) inserting an outer shield integrated optical visualization instrument,
- f) inserting an inner shield through the outer shield so that a distal end portion of the inner shield extends to the disc, the inner shield having an outer surface, 20
- g) contacting the outer surface of the inner shield to a nerve root to shield the nerve root,
- h) with the nerve root shielded, conducting a surgical operation through the access tube.

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